Universidad de Ibagué

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ECG Heartbeat Detector with Operational Amplifiers

Precision Rectifier, Peak Detector and Schmitt Comparator

Laboratory Report

Nonlinear Circuits

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Abstract

A heartbeat detector that delivers one clean rectangular pulse for every QRS complex of an ECG-like signal was designed, simulated and built. The circuit uses six operational amplifiers powered from $\pm 12\,\mathrm{V}$ and consists of five stages: a non-inverting amplifier with a gain of 4.9, a full-wave precision rectifier, a buffered peak detector with a discharge time constant $\tau = 100\,\mathrm{ms}$, a Schmitt comparator with thresholds $V_{TH} = 1.01\,\mathrm{V}$ and $V_{TL} = 0.52\,\mathrm{V}$, and an LED indicator. Each stage was sized from the signal specifications (amplitude of 0.5 to $2\,\mathrm{V_{pp}}$, width of 75 ms, rate of 40 to 150 beats per minute) and verified with a behavioral simulation in Python and with Proteus. In simulation the circuit detected every beat within the design range for both polarities; the only failing case, 180 bpm at $2\,\mathrm{V_{pp}}$, matches the theoretical limit of 163 bpm predicted by the discharge analysis. The breadboard build with two LM324N chips was tested with the cardiac waveform of the Agilent 33120A generator at 1.2 Hz: it delivered one pulse per beat (1.199 Hz measured at the output), with peaks of 4.23 V at V_r and 4.32 V at V_{env}, and it still detected every beat with a $4\,\mathrm{V_{pp}}$ input, twice the design maximum.

Keywords: precision rectifier, peak detector, Schmitt comparator, hysteresis, ECG, operational amplifier.

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1 Introduction

The electrocardiogram (ECG) records the electrical activity of the heart. Each heartbeat contains three events: the P wave (atrial depolarization), the QRS complex (ventricular depolarization) and the T wave (ventricular repolarization) [4]. The QRS consists of a small negative deflection (Q), a large peak (R) and another negative deflection (S), and lasts 80 to 120 ms in an adult. Because it is the part of the ECG with the largest amplitude and slope, and it appears only once per beat, detecting it is the first step of any heart-rate monitor [5].

Linear circuits alone are not enough to detect the QRS: the R wave can be positive or negative depending on the lead, the complex has several lobes that could be counted as several beats, and the output must be a digital signal (beat or no beat) that can be counted. For this reason the detector is built from nonlinear blocks: a precision rectifier that removes the dependence on polarity, a peak detector that merges the lobes into a single event, and a comparator with hysteresis that turns the envelope into a clean pulse.

This report presents the complete design of that circuit: the derivation of each equation, the choice of each component, the verification by simulation and the measurement procedure in the laboratory.

2 Problem Statement

> *Design and implement a heartbeat detector based on operational amplifiers powered from* $\pm 12\,\mathrm{V}$*, able to identify every QRS complex of an ideal ECG-like signal of 0.5 to* $2\,\mathrm{V_{pp}}$*, regardless of its polarity or signal type, delivering one clean rectangular pulse per beat over a range of 40 to 150 beats per minute.*

The assignment only sets the conditions of the signal and of the circuit build; all other values (gain, time constant, thresholds) are outcomes of the design. The starting data are:

- LM324 or TL074 op-amps powered from $\pm 12\,\mathrm{V}$, which saturate at $\pm V_{sat} \approx \pm 10.5\,\mathrm{V}$.
- Test signal (function generator): a 1.2 Hz (72 bpm) rectangular pulse, 75 ms wide, of 0.5 to $2\,\mathrm{V_{pp}}$ with its low level at 0 V, as a model of the QRS; and a 0.5 Hz sine wave of 0.5 to $2\,\mathrm{V_{pp}}$.
- Operating range: 40 to 150 bpm, with a positive or negative R wave.
- Output: one pulse per beat, visible on an LED.

3 Objectives

3.1 General objective

Design, simulate and implement an ECG heartbeat detector with operational amplifiers that delivers one pulse for every QRS complex, regardless of its polarity.

3.2 Specific objectives

- Design a full-wave precision rectifier and analyze its operation in each half-cycle.
- Size a peak detector whose time constant merges the lobes of the QRS and recovers before the next beat.
- Design a Schmitt comparator whose thresholds and hysteresis prevent false triggering.
- Verify the design with a behavioral simulation and with Proteus, using pulses, sine waves and a synthetic heartbeat.
- Measure the assembled circuit and compare the results with the calculated and simulated values.

4 Theoretical Background

4.1 Operational amplifier model

Two rules are used with the ideal op-amp: (1) no current flows into its inputs, and (2) with negative feedback the output adjusts itself so that both inputs sit at the same voltage ($V_- = V_+$), which is called a *virtual short*. If one input is grounded, the other one sits at 0 V without being connected to ground: it is a *virtual ground*.

With positive feedback, or with no feedback, the virtual short does not hold: the op-amp amplifies the difference between its inputs by its open-loop gain ($A_{ol} \approx 200\,000$) and its output saturates at $\pm V_{sat}$. A difference of only $21\,\mathrm{V}/200\,000 \approx 0.1\,\mathrm{mV}$ is enough to swing the whole output range, so in practice it behaves as a switch.

4.2 Precision rectifier

A silicon diode does not conduct until its voltage exceeds about 0.6 V, so a passive rectifier loses that voltage and does not respond to small signals. If the diode is placed inside the feedback loop of an op-amp, the op-amp output shifts as much as needed to compensate for the diode drop, and the dead zone shrinks to V_D/A_{ol}, on the order of microvolts.

4.3 Peak detector

A peak detector charges a capacitor through a diode up to the maximum value of the signal. The diode acts as a one-way valve: it lets charge in when the signal exceeds the capacitor voltage and blocks it when the signal falls. A parallel resistor slowly discharges the capacitor with a time constant $\tau = RC$, so the output follows the envelope of the signal.

4.4 Comparator with hysteresis (Schmitt)

A simple comparator changes state when the input crosses a threshold; if the input is noisy near the threshold, the output bounces several times. The Schmitt comparator uses positive feedback to obtain two thresholds: a rising one (V_{TH}) and a falling one ($V_{TL} < V_{TH}$). The difference $\Delta V = V_{TH} - V_{TL}$ is the hysteresis, and noise smaller than it does not cause bouncing.

5 Materials and Equipment

5.1 Equipment

- Agilent 33120A function/arbitrary waveform generator (15 MHz), with a built-in cardiac waveform (CARDIAC).
- GW Instek GDS-1102A-U digital oscilloscope (100 MHz, two channels).
- GW Instek dual power supply in series mode, set to ±12 V.
- Digital multimeter.
- Proteus 8 (simulation) and Python with NumPy and Matplotlib (behavioral simulation).

5.2 Bill of materials

Resistors R2 to R6 must be 1 % metal film; the rest can be 5 %.

Table 1: Bill of materials.

Ref.	Component	Value	Qty.	Block
U1–U6	LM324N quad op-amp (or TL074) + DIP-14 socket	—	2	All
R0	Resistor	100 kΩ	1	Input
Rf1	Resistor	39 kΩ	1	Amplifier
Rg1, R2, R3, R5, R8, R10	Resistor	10 kΩ	6	Various
R4, R6	Resistor 1 %	20 kΩ	2	Summer
R7	Resistor	100 Ω	1	Detector
R9	Resistor	100 kΩ	1	Detector
R11	Resistor	430 kΩ	1	Schmitt
R12	Resistor	15 kΩ	1	V_{ref}
R13	Resistor	1 kΩ	1	V_{ref}
R14	Resistor	1 kΩ	1	LED
D1–D5	1N4148 signal diode	—	5	Rectifier, detector, LED
LED1	5 mm LED (blue in the build)	—	1	Indicator
C1	Polyester capacitor	1 µF	1	Detector

Ref.	Component	Value	Qty.	Block
C2–C5	Ceramic decoupling	100 nF	4	Power supply
C6, C7	Electrolytic	10 µF, 25 V	2	Power supply

The two spare sections of the second IC are wired as followers with the non-inverting input grounded, so that they do not oscillate.

6 Design and Calculations

6.1 Specifications

Table 2: Starting data (taken from the assignment).

Parameter	Value
Power supply	$\pm 12\,\mathrm{V}$; saturation $V_{sat} \approx 10.5\,\mathrm{V}$ (TL074)
Input amplitude	0.5 to $2\,\mathrm{V_{pp}}$; pulse with low level at $0\,\mathrm{V} \Rightarrow$ peak of 0.5 to 2 V
Pulse width (QRS)	$w = 75\,\mathrm{ms}$
Heart rate	40 to 150 bpm (period of 1.5 s to 400 ms)
Polarity	Positive or negative R wave
Output	One pulse per beat, with LED indicator

The calculations use $V_{sat} \approx 10.5\,\mathrm{V}$ (TL074 at $\pm 12\,\mathrm{V}$). With an LM324 or LM741 the values change by only a few millivolts (section 10.2).

Design order. The stages are presented in the order in which the signal goes through them, but they are sized in the order in which they depend on each other, because each one uses the result of the previous one:

1. **Gain (stage 1):** depends only on saturation and on the maximum input. It sets the range of peaks seen by the other stages.
2. **Rectifier (stage 2):** unity gain; only resistor ratios need to be set.
3. **Schmitt thresholds (stage 4):** depend on the minimum peak left by stage 1 and on noise.
4. **Time constant (stage 3):** depends on the maximum peak (stage 1), on V_{TL} (stage 4) and on the QRS timing.
5. **LED indicator (stage 5):** depends only on V_{sat}.

6.2 Block diagram

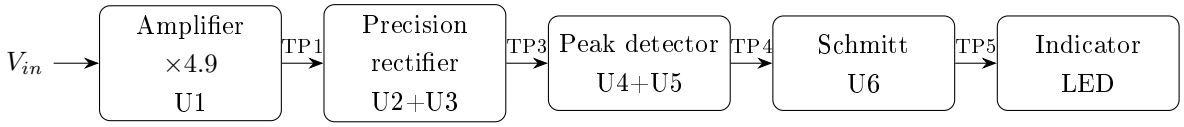


Figure 1: Block diagram of the detector and test points.

6.3 Schematic

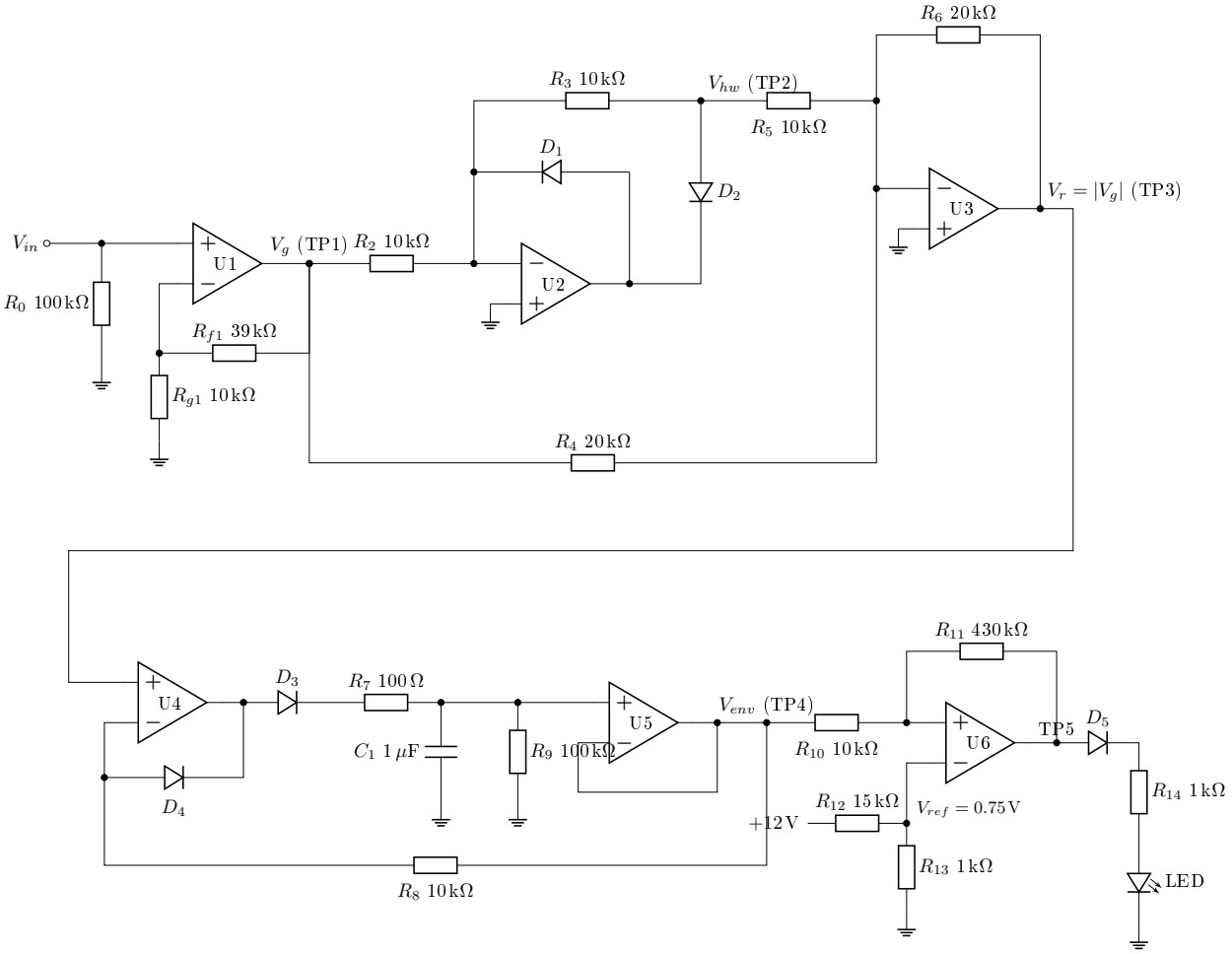


Figure 2: Complete schematic. The ±12 V supply and the 100 nF decoupling capacitors are not drawn.

The signal runs along the first row from left to right and moves to the second row at TP3. The connections of each op-amp are:

1. **U1 (amplifier):** (+) ← input, with R0 to ground; (−) ← Rg1 to ground and Rf1 from the output. Output = V_g (TP1).

2. **U2 (half-wave):** (+) to ground; (−) ← V_g through R2. D1: anode at the U2 output, cathode

at (−). D2: cathode at the U2 output, anode at node V_{hw}. R3 from (−) to V_{hw} (TP2).

3. **U3 (summer):** (+) to ground; (−) ← V_g through R4 and ← V_{hw} through R5; R6 from (−) to the output. Output = $V_r = |V_g|$ (TP3).

4. **U4 (detector):** (+) ← V_r. Output → D3 → R7 → C1 node (C1 and R9 in parallel to ground). (−) ← V_{env} through R8. D4: anode at (−), cathode at the U4 output.

5. **U5 (buffer):** follower; (+) ← C1 node, (−) tied to the output. Output = V_{env} (TP4).

6. **U6 (Schmitt):** (+) ← V_{env} through R10 and ← U6 output through R11. (−) ← V_{ref} (divider: R12 from +12 V, R13 to ground). Output = TP5.

7. **Indicator:** TP5 → D5 → R14 → LED → ground.

6.4 Stage 1: non-inverting amplifier (U1)

This is the first stage to be designed: it brings the input signal to a comfortable level for the rest of the circuit, moving it away from noise and offsets without reaching saturation.

Derivation. The non-inverting input is connected to the signal, so $V_+ = V_{in}$. The inverting input takes a fraction of the output through the Rf1–Rg1 divider:

$$V_- = V_g \frac{R_{g1}}{R_{g1} + R_{f1}} \tag{1}$$

Setting $V_- = V_+ = V_{in}$ and solving:

$$V_g = V_{in}\left(1 + \frac{R_{f1}}{R_{g1}}\right) = V_{in}\left(1 + \frac{39\,\text{k}\Omega}{10\,\text{k}\Omega}\right) = 4.9\,V_{in} \tag{2}$$

Choice of gain. At the start of the design only the input (peaks of 0.5 to 2 V) and the op-amp saturation ($V_{sat} \approx 10.5$ V) are known; the comparator thresholds do not exist yet. The input varies over a 4:1 ratio, so it is not possible to place both the large and the small signal at an ideal level at the same time. The criterion is to **amplify as much as possible without saturating the largest signal**, so that the smallest one ends up as far as possible from noise and offsets.

1. **Saturation limit.** The maximum input is the $2\,V_{pp}$ pulse with its low level at 0 V, whose peak is 2 V:

$$G \cdot 2\,\text{V} \le V_{sat} \;\Rightarrow\; G_{max} = \frac{10.5}{2} = 5.25 \tag{3}$$

   Above this value the pulse is clipped and the peak detector no longer sees the real amplitude.

2. **Standard value.** With $R_{g1} = 10\,\text{k}\Omega$, $R_{f1} = (G-1)R_{g1} \le 42.5\,\text{k}\Omega$. In the E12 series:

   - $R_{f1} = 39\,\text{k}\Omega \Rightarrow G = 4.9$, 7 % below saturation.

- $R_{f1} = 47\,\text{k}\Omega \Rightarrow G = 5.7$, which saturates.

$G = 4.9$ is chosen, the largest standard gain that does not saturate.

3. **Resulting range.** The amplified signal has peaks of

$$V_{p,min} = 4.9 \cdot 0.5 = 2.45\,\text{V}, \qquad V_{p,max} = 4.9 \cdot 2 = 9.8\,\text{V} \tag{4}$$

These values are the starting data for the following stages.

4. **Requirement passed on to the comparator.** To detect the smallest signal with a margin of 2 against noise and tolerances, the Schmitt trigger threshold must satisfy

$$V_{TH} \leq \frac{V_{p,min}}{2} = \frac{2.45}{2} = 1.22\,\text{V} \tag{5}$$

This condition is used in stage 4.

Figure 3 shows the hand-worked reasoning behind this criterion: the saturation limit ($G \leq 5.25$), the choice of $G = 4.9$ with margin, and the check of the maximum (9.8 V) and minimum (2.45 V) peaks against V_{sat}.

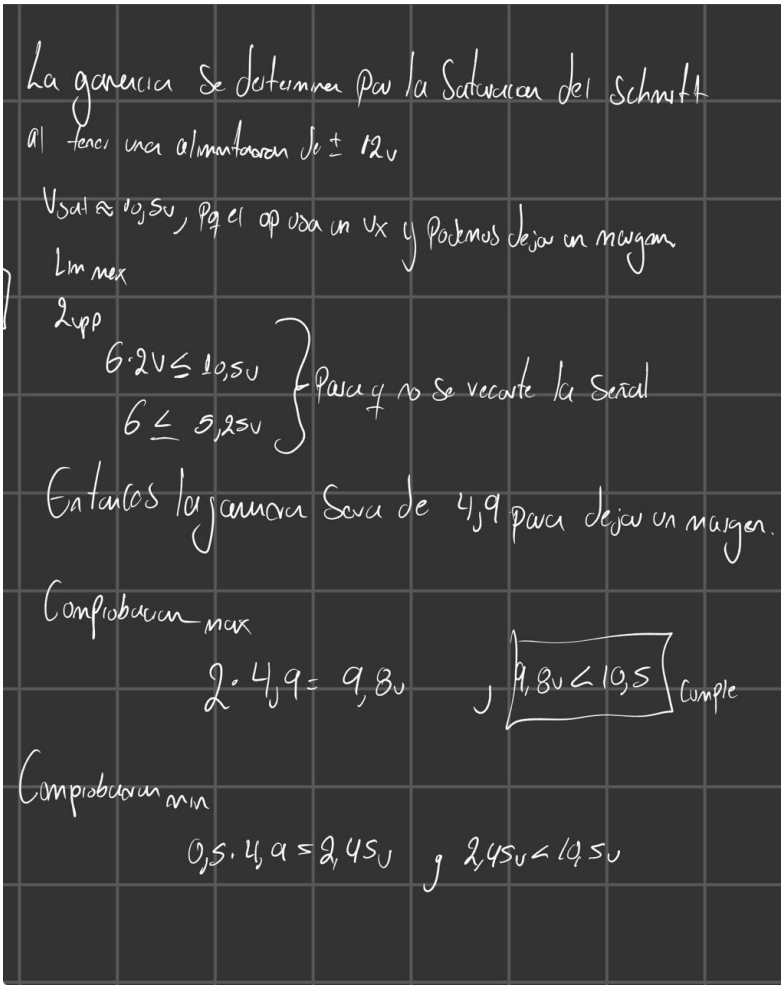


Figure 3: Hand calculation: choice of the non-inverting amplifier gain from saturation (handwritten, in Spanish).

Verification. With the Schmitt already designed ($V_{TH} = 1.01\,\text{V}$, stage 4), we check that the gain falls within the window imposed by both stages:

$$\underbrace{\frac{2 \cdot 1.01}{0.5}}_{4.04} \leq G = 4.9 \leq \underbrace{\frac{10.5}{2}}_{5.25} \tag{6}$$

Non-idealities. With a finite open-loop gain,

$$G_{real} = \frac{G}{1 + G/A_{ol}} = \frac{4.9}{1 + 4.9/200\,000} \approx 4.8999 \tag{7}$$

with an error of 0.0025 %. The bandwidth is $f_{-3dB} = GBW/G$: 612 kHz with the TL074 and 204 kHz with the LM324. The 0 to 9.8 V edge takes 0.75 µs with the TL074 and 20 µs with the LM324, negligible compared with 75 ms. The generator (50 Ω) sees R0 = 100 kΩ: the attenuation is 0.9995.

Hand calculations. Figure 4 reproduces the calculation done by hand: with $R_{g1} = 10\,\text{k}\Omega$ and $A_v = 4.9$, solving gives $R_{f1} = (A_v - 1)R_{g1} = 39\,\text{k}\Omega$.

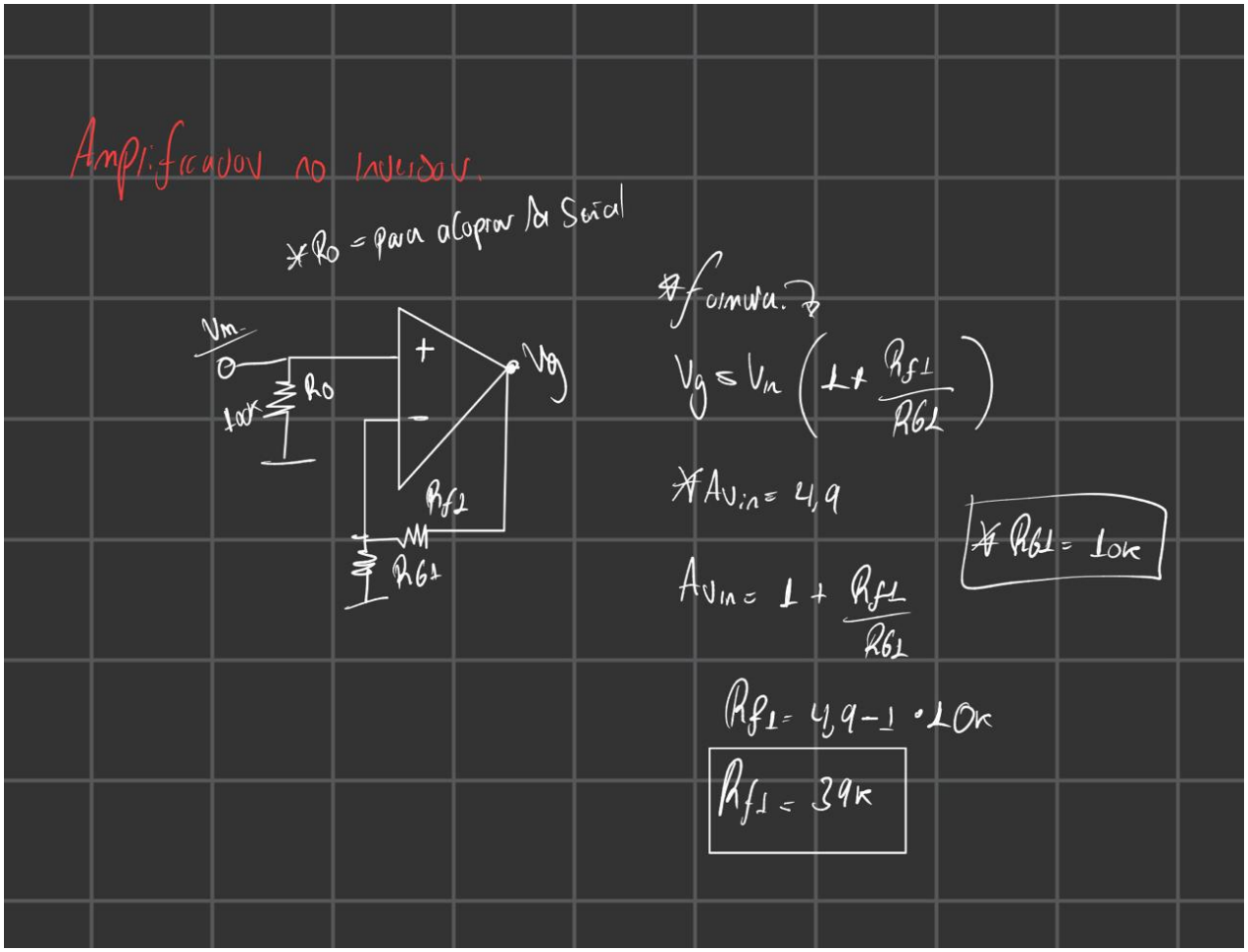


Figure 4: Hand calculations: non-inverting amplifier (U1) (handwritten, in Spanish).

6.5 Stage 2: full-wave precision rectifier (U2 + U3)

This stage delivers $V_r = |V_g|$ with unity gain for both polarities.

Positive half-cycle ($V_g > 0$). The (+) input of U2 is grounded, so its (−) input is a virtual ground. A current $i = V_g/R_2$ enters through R2; since it cannot flow into the op-amp, it continues through R3 toward node V_{hw}. The U2 output goes down, D2 conducts and D1 is reverse-biased. Applying the current law at the inverting input:

$$\frac{V_g - 0}{R_2} = \frac{0 - V_{hw}}{R_3} \;\Rightarrow\; V_{hw} = -\frac{R_3}{R_2}V_g = -V_g \tag{8}$$

The U2 output sits one diode drop below: $V_{hw} - 0.6\,\text{V}$. That drop does not appear at V_{hw} because the signal is taken after D2, inside the loop.

Negative half-cycle ($V_g < 0$). The current through R2 leaves the node; the U2 output rises until D1 conducts and settles at $\approx +0.6\,\mathrm{V}$, almost independent of V_g. D2 is reverse-biased, no current flows through R3 and R5, and $V_{hw} = 0$.

Summer (U3). With a virtual ground at its $(-)$ input, the current through R6 is the sum of the currents arriving through R4 and R5:

$$\frac{V_g}{R_4} + \frac{V_{hw}}{R_5} = -\frac{V_r}{R_6} \;\Rightarrow\; V_r = -\left(\frac{R_6}{R_4}V_g + \frac{R_6}{R_5}V_{hw}\right) = -\left(V_g + 2V_{hw}\right) \tag{9}$$

Substituting each half-cycle:

$$V_g > 0:\; V_r = -(V_g - 2V_g) = +V_g \qquad V_g < 0:\; V_r = -(V_g + 0) = -V_g \tag{10}$$

In both cases $V_r = |V_g|$. The path through R4 carries the full signal (before the rectifier) with weight 1; the path through R5 carries the half-wave signal with weight 2, which in the positive half-cycle cancels the wrong-sign term and adds it back with the correct sign.

Table 3: Voltages and diode states with $V_g = \pm 4\,\mathrm{V}$.

	$V_g = +4\,\mathrm{V}$	$V_g = -4\,\mathrm{V}$
U2 output	$-4.6\,\mathrm{V}$	$\approx +0.62\,\mathrm{V}$
D1 / D2	Off / on	On / off
V_{hw} (TP2)	$-4\,\mathrm{V}$	$0\,\mathrm{V}$
Current through D2	$0.8\,\mathrm{mA}$ (R3 + R5)	0
V_r (TP3)	$+4\,\mathrm{V}$	$+4\,\mathrm{V}$

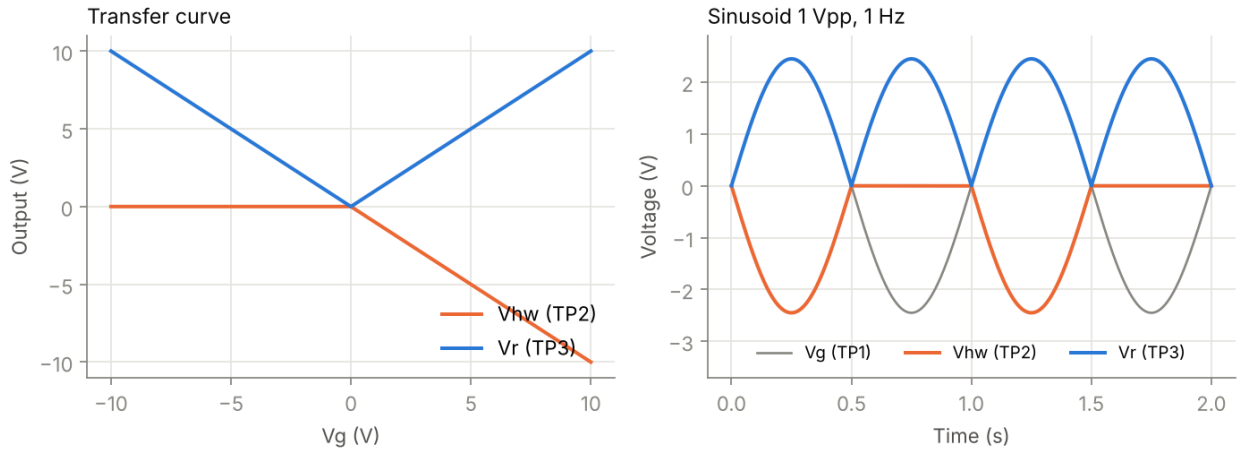


Figure 5: Rectifier: transfer curve (left) and waveforms with a sine wave (right).

Dead zone and transition. For the diode to conduct, an input difference of only

$$V_{dead\,zone} = \frac{V_D}{A_{ol}} = \frac{0.6\,\mathrm{V}}{200\,000} = 3\,\mu\mathrm{V} \tag{11}$$

is needed. D1 keeps U2 from saturating: without it, in the negative half-cycle the loop would be open and the output would rise to $+10.5\,\mathrm{V}$. With D1 the output only jumps $1.2\,\mathrm{V}$ at each zero crossing, which takes $0.09\,\mu\mathrm{s}$ with the TL074 and $2.4\,\mu\mathrm{s}$ with the LM324.

Sensitivity to tolerances. Without assuming equal resistors:

$$G_+ = \frac{R_6}{R_5}\cdot\frac{R_3}{R_2} - \frac{R_6}{R_4}, \qquad G_- = \frac{R_6}{R_4} \tag{12}$$

G_+ is the subtraction $2-1$, which carries the full error of each term. Evaluating the 32 combinations of $\pm$tolerance and a Monte Carlo simulation of 20 000 circuits:

Table 4: Difference between the gains of each half-cycle versus tolerance.

Resistors	Worst case $\lvert G_+ - G_-\rvert / G_-$	95th percentile
1 %	8.2 %	2.6 %
5 %	44 %	13 %

Hand calculations. Figures 6 and 7 show the hand analysis of the half-wave rectifier in each half-cycle (with the example $V_g = \pm 4\,\mathrm{V}$) and of the summer, from which the conditions $R_3 = R_2$, $R_6 = R_4$ and $R_6 = 2R_5$ follow.

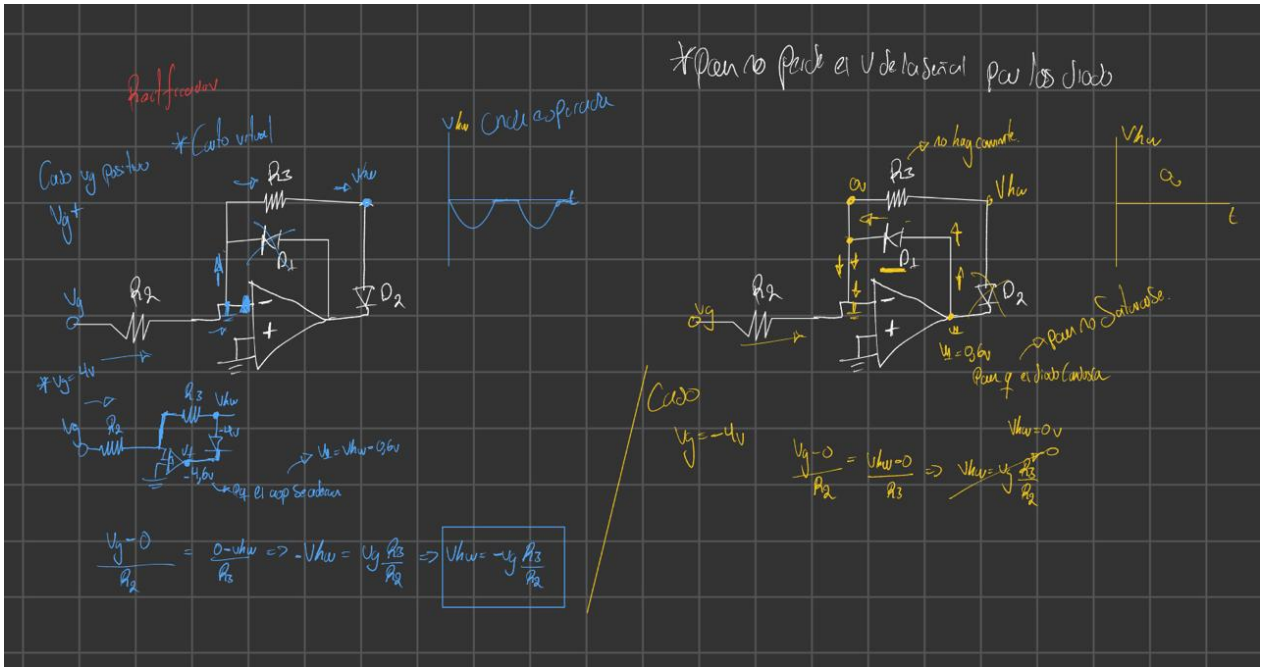


Figure 6: Hand calculations: half-wave rectifier (U2) with positive and negative V_g (handwritten, in Spanish).

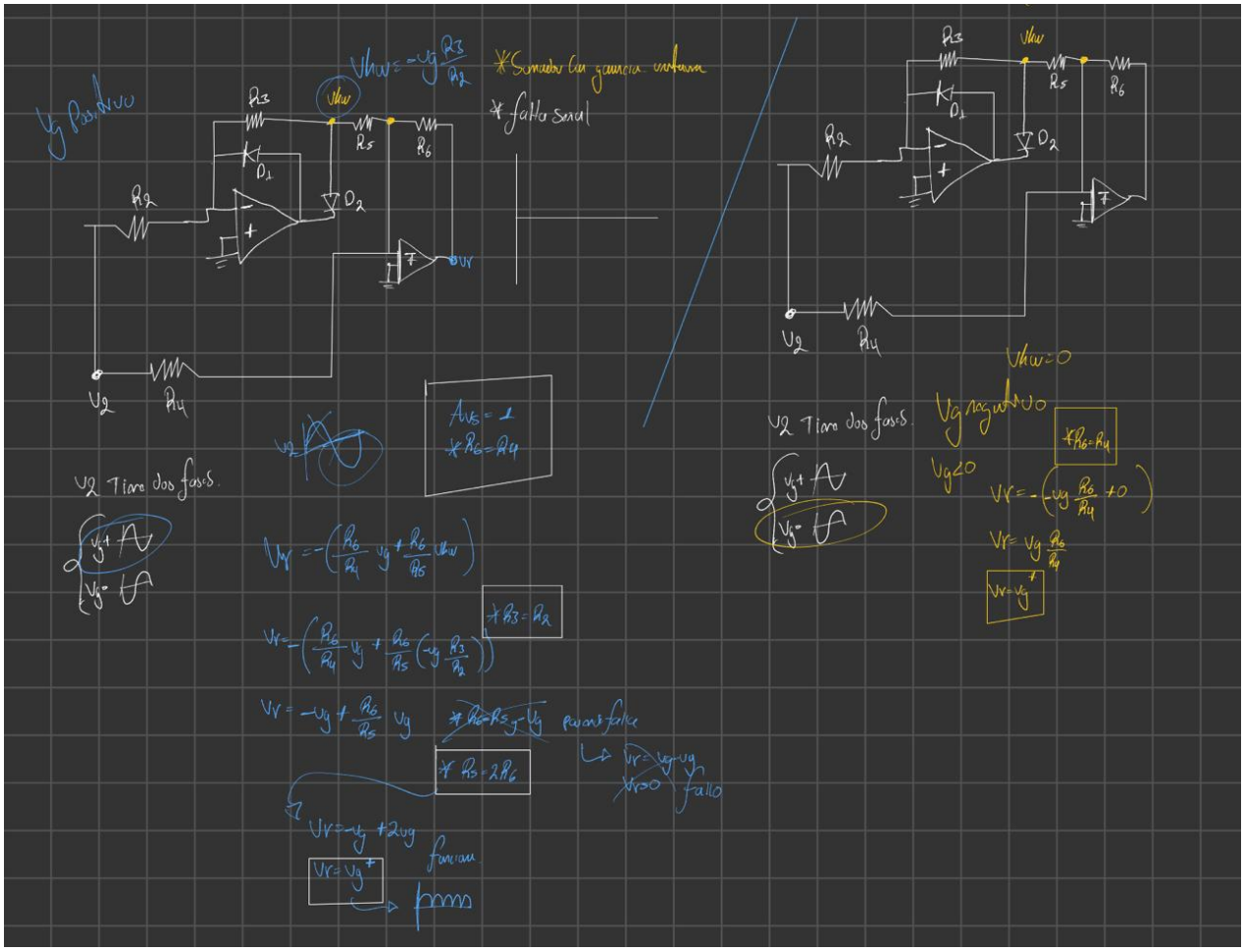


Figure 7: Hand calculations: inverting summer (U3) and conditions on R4, R5 and R6 (handwritten, in Spanish).

6.6 Stage 3: peak detector (U4 + U5, C1, R9)

The detector follows the rise of V_r in less than 1 ms, holds it for the duration of the pulse and then discharges C1 with $\tau = 100\,\text{ms}$.

Table 5: Starting data for the peak detector.

Item	Value	Source
Peak of V_r	2.45 to 9.8 V	Input of 0.5 to 2 V_{pp} times the gain of 4.9
Pulse width (w)	75 ms	Test pulse, similar to a QRS
Spacing between QRS lobes	≈ 20 ms	Typical time between the R and S waves
Maximum droop between lobes	20 %	Design criterion
Maximum heart rate	150 bpm ($T = 400$ ms)	Design range
Release threshold	$V_{TL} = 0.52$ V	Stage 4
Maximum op-amp current	≈ 20 mA	Datasheet
Bias current	≈ 80 nA	741 datasheet (worst case)

6.6.1 Operation

Charging ($V_r > V_{env}$). U4 compares V_r (+ input) with V_{env}, which reaches it through R8 from buffer U5 (− input). If V_r is larger, the U4 output rises, D3 conducts and C1 charges through R7 until $V_{env} = V_r$. Since R8 takes the voltage after D3, the diode drop stays inside the loop: the U4 output settles ≈ 0.5 V above V_{env} and V_{env} comes out equal to V_r. Charging is limited by the maximum op-amp current:

$$\frac{dV_{env}}{dt} = \frac{I_{max}}{C_1} = \frac{20\,\text{mA}}{1\,\mu\text{F}} = 20\,\text{V/ms} \tag{13}$$

Reaching 9.8 V takes ≈ 0.5 ms.

Discharging ($V_r < V_{env}$). When the pulse ends, V_r drops to zero immediately, but C1 cannot change its voltage instantly. D3 is reverse-biased and the buffer input draws no current, so C1 discharges only through R9:

$$C_1 \frac{dV_{env}}{dt} = -\frac{V_{env}}{R_9} \;\Rightarrow\; V_{env}(t) = V_p\, e^{-t/\tau}, \qquad \tau = R_9 C_1 \tag{14}$$

D4 closes the loop of U4: its (−) input sits at $V_r = 0$ and its output at ≈ -0.6 V, without saturating. The difference between V_{env} and that input appears across R8 (0.4 mA with $V_{env} = 4$ V), and that current is supplied by the buffer, not by C1.

Role of the buffer. U5 is a follower ($V_{out} = V_{in}$, input impedance on the order of $10^{11}\,\Omega$). Without it, C1 would also discharge through R8 and τ would drop to $(R_9 \parallel R_8)C_1 \approx 9$ ms; the envelope would fall between the QRS lobes and a single beat would produce several pulses.

Time to release the comparator. From $V_p\, e^{-t/\tau} = V_{TL}$:

$$t_{desc} = \tau \, \ln \frac{V_p}{V_{TL}} \tag{15}$$

Table 6: Discharge time and output pulse width.

Input	V_p	t_{desc}	Output width $(w + t_{desc})$
0.5 V_{pp}	2.45 V	154 ms	229 ms
1 V_{pp}	4.90 V	224 ms	299 ms
2 V_{pp}	9.80 V	293 ms	368 ms

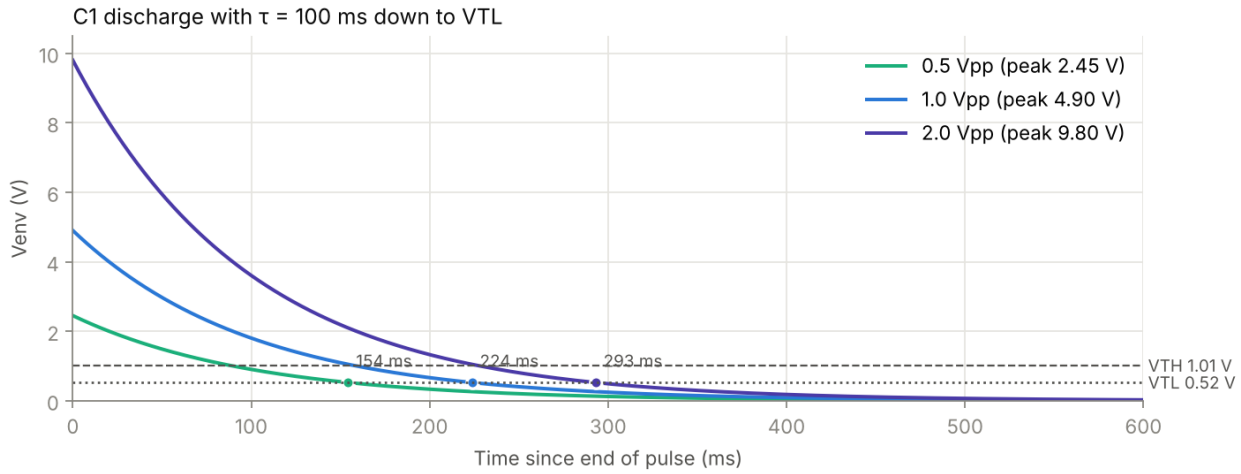


Figure 8: Discharge of C1 down to V_{TL} for the three amplitudes.

6.6.2 Component design

Lower limit of τ. The QRS is not a single pulse: its lobes (Q, R and S) are about 20 ms apart and the whole complex lasts 80 to 120 ms in an adult [4]. Between lobes the envelope must not lose more than 20 % (design criterion), so that the comparator is not released in the middle of the complex and does not count one beat twice:

$$e^{-20\,\mathrm{ms}/\tau} > 0.8 \;\Rightarrow\; \tau > \frac{20\,\mathrm{ms}}{\ln(1/0.8)} \approx 90\,\mathrm{ms} \tag{16}$$

Discharge condition. The condition for the upper limit is derived by following one complete beat:

1. **Charge.** The pulse arrives and C1 charges to the peak V_p in ≈ 0.1 ms ($R_7 C_1$). V_{env} exceeds V_{TH} and the Schmitt output goes high.

2. **Hold.** For the duration of the pulse (w), U4 holds C1 at V_p.

3. **Discharge.** When the pulse ends, D3 turns off and C1 discharges only through R9, following the natural response of an RC circuit [1, 3]: $V_{env}(t) = V_p\, e^{-t/\tau}$.

4. **Release.** The Schmitt returns low when $V_{env} = V_{TL}$ [1], which happens after the discharge time $t_{desc} = \tau \ln(V_p/V_{TL})$ of equation (15).

From the start of one beat until the comparator is ready for the next one, $w + t_{desc}$ elapses. If the next beat arrives earlier, V_{env} is still above V_{TL}, the output does not go low and the two beats are counted as one. The condition for detecting each beat separately is therefore

$$w + \tau \ln \frac{V_p}{V_{TL}} < T = \frac{60}{HR} \tag{17}$$

This time plays the same role as the *refractory period* of software QRS detectors, which ignore any event for about 200 ms after each beat [5]. Here the refractory period is not programmed: it is set by τ, V_p and V_{TL}.

Upper limit of τ. The worst case is the highest peak (9.8 V, which takes longest to discharge) at the highest rate in the assignment (150 bpm, $T = 400$ ms). With $V_{TL} = 0.52$ V (stage 4) in equation (17):

$$75\,\text{ms} + \tau \ln \frac{9.8}{0.52} < 400\,\text{ms} \;\Rightarrow\; \tau < \frac{325\,\text{ms}}{2.94} \approx 110\,\text{ms} \tag{18}$$

The midpoint of the 90 ms to 110 ms window is chosen, $\tau = 100$ ms.

Maximum heart rate. Solving equation (17) for HR gives the maximum rate the circuit can count for any amplitude:

$$HR_{max} = \frac{60}{w + \tau \ln(V_p/V_{TL})}\ \text{bpm} \tag{19}$$

Table 7: Maximum heart rate with $\tau = 100$ ms, $w = 75$ ms and $V_{TL} = 0.523$ V.

Input	V_p	V_p/V_{TL}	$t_{desc} = \tau \ln(V_p/V_{TL})$	$w + t_{desc}$	HR_{max}
0.5 V_{pp}	2.45 V	4.68	154 ms	229 ms	262 bpm
1 V_{pp}	4.9 V	9.37	224 ms	299 ms	201 bpm
2 V_{pp}	9.8 V	18.7	293 ms	368 ms	163 bpm

Example with 2 V_{pp}: $t_{desc} = 0.1 \ln(18.7) = 0.293$ s, $w + t_{desc} = 0.368$ s and $HR_{max} = 60/0.368 = 163$ bpm. The larger the peak, the longer C1 takes to fall to V_{TL}, which is why the maximum rate decreases with amplitude. In every case it stays above the 150 bpm required by the assignment.

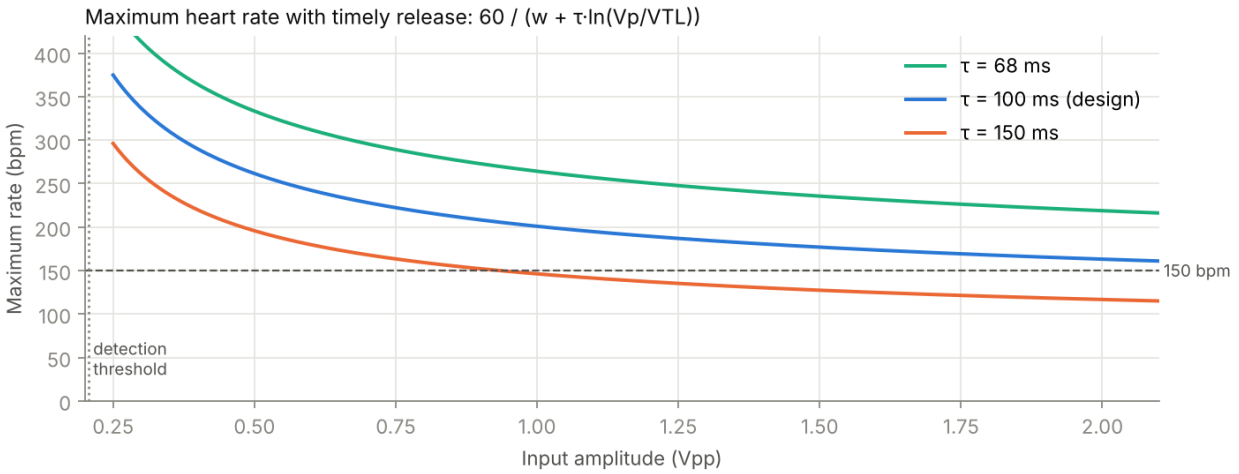


Figure 9: Maximum heart rate versus input amplitude for three values of τ.

C1 and R9. The 100 ms product allows several combinations:

Table 8: Options for obtaining $\tau = 100\,\text{ms}$.

C1	R9	Outcome
10 µF	10 kΩ	Rejected: an electrolytic has µA leakage and ±20 % tolerance
1 µF	100 kΩ	Chosen: stable polyester; bias error 80 nA × 100 kΩ = 8 mV
100 nF	1 MΩ	Rejected: bias error of 80 mV, 15 % of V_{TL}

R7 and R8. R7 must give a charge much faster than the rise of the QRS ($\tau_c = R_7 C_1 \leq 1\,\text{ms}$, i.e. $R_7 \leq 1\,\text{k}\Omega$) and isolate the op-amp from the capacitive load (more than $\approx 50\,\Omega$); 100 Ω was chosen. During discharge, R8 carries $(V_{env} - V_r)/R_8$, at most 0.98 mA with 10 kΩ; with MΩ resistors the bias current would create errors of tens of mV. Its useful range is 1 kΩ to 100 kΩ, and 10 kΩ was chosen.

Table 9: Summary of the detector sizing.

Component	Governing condition	Valid range	Value
$\tau = R_9 C_1$	Merge the QRS and release at 150 bpm	90–110 ms	100 ms
C1	Stable film	—	1 µF
R9	τ / C_1	—	100 kΩ
R7	Fast charge and stability	100 Ω–1 kΩ	100 Ω
R8	Buffer current and bias	1 kΩ–100 kΩ	10 kΩ

6.6.3 Behavior with a sine wave

With $V_r = A|\sin \omega t|$ the envelope follows the signal while the signal falls more slowly than the exponential, and separates from it when $\tan\varphi < \omega\tau$. At the zero crossing it is approximately

$$V_{env,min} \approx A \sin\varphi_0 \, e^{-\varphi_0/(\omega\tau)}, \qquad \varphi_0 = \arctan(\omega\tau) \tag{20}$$

With $A = 4.9\,\mathrm{V}$: at $0.5\,\mathrm{Hz}$, $V_{env,min} \approx 0.56\,\mathrm{V}$ (the simulation gives $0.42\,\mathrm{V}$) and two pulses per cycle are seen; at $1\,\mathrm{Hz}$, $V_{env,min} \approx 1.07\,\mathrm{V} > V_{TL}$ and the comparator stays high. This is why the sine-wave test is done at $0.5\,\mathrm{Hz}$.

Hand calculations. Figure 10 collects the hand analysis of charging and discharging, the equation for $V_{env}(t)$ and the calculation of $C_1 = 1\,\mu\mathrm{F}$ and $R_9 = 100\,\mathrm{k}\Omega$.

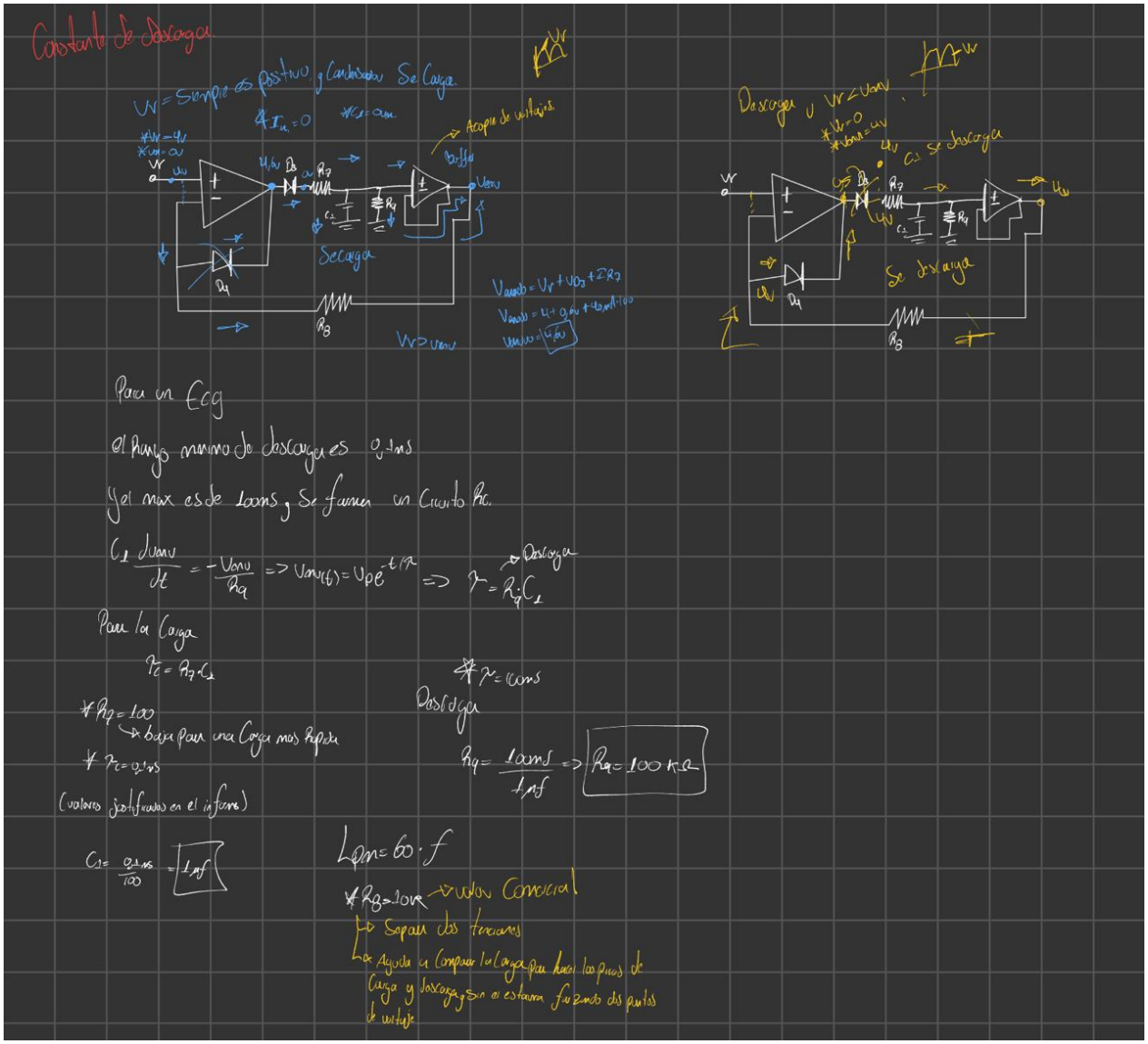


Figure 10: Hand calculations: peak detector (U4, U5), charging, discharging and time constant (handwritten, in Spanish).

6.7 Stage 4: non-inverting Schmitt comparator (U6)

Choice of thresholds. This stage is designed right after stage 1, because its thresholds depend on the minimum peak of the amplified signal ($V_{p,min} = 2.45\,\mathrm{V}$) and stage 3 needs to know V_{TL}. The thresholds are chosen with margin relative to the signals:

- $V_{TH} \approx 1\,\mathrm{V}$, which meets condition (5) ($V_{TH} \leq 1.22\,\mathrm{V}$): it is at half or less of the minimum pulse peak (2.45 V) and below the peak of the $0.5\,\mathrm{V_{pp}}$ sine wave (1.22 V).
- $V_{TL} \approx 0.5\,\mathrm{V}$, well above the accumulated offset on V_{env} (tens of mV) and below the T wave.
- $\Delta V \approx 0.5\,\mathrm{V}$, about ten times the expected noise on the envelope.

Equations. U6 has no negative feedback: its output is always at $\pm V_{sat}$ and there is no virtual short. By superposition, with $k = R_{10}/R_{11}$:

$$V_+ = V_{env}\,\frac{R_{11}}{R_{10}+R_{11}} + V_o\,\frac{R_{10}}{R_{10}+R_{11}} \tag{21}$$

The output switches when $V_+ = V_{ref}$:

$$V_{env} = V_{ref}(1+k) - V_o\,k \tag{22}$$

With $V_o = -V_{sat}$ (before the beat) and $V_o = +V_{sat}$ (during the beat):

$$V_{TH} = V_{ref}(1+k) + V_{sat}\,k, \qquad V_{TL} = V_{ref}(1+k) - V_{sat}\,k \tag{23}$$

$$\Delta V = V_{TH} - V_{TL} = 2V_{sat}\,k, \qquad V_c = \frac{V_{TH}+V_{TL}}{2} = V_{ref}(1+k) \tag{24}$$

Sizing.

1. $k = \Delta V/(2V_{sat}) = 0.5/21 = 0.0238$. With $R_{10} = 10\,\text{k}\Omega$: $R_{11} = R_{10}/k = 420\,\text{k}\Omega \to 430\,\text{k}\Omega$ (E24), $k = 0.02326$.
2. $V_{ref} = V_c/(1+k) \approx 0.75\,\text{V}$. With the divider from $+12\,\text{V}$: $R_{13}/(R_{12}+R_{13}) = 1/16$, i.e. $R_{12} = 15\,\text{k}\Omega$ and $R_{13} = 1\,\text{k}\Omega$.

With the standard values:

$$V_{TH} = 0.750(1.02326) + 10.5(0.02326) = 0.767 + 0.244 = 1.012\,\text{V} \tag{25}$$

$$V_{TL} = 0.767 - 0.244 = 0.523\,\text{V}, \qquad \Delta V = 0.488\,\text{V} \tag{26}$$

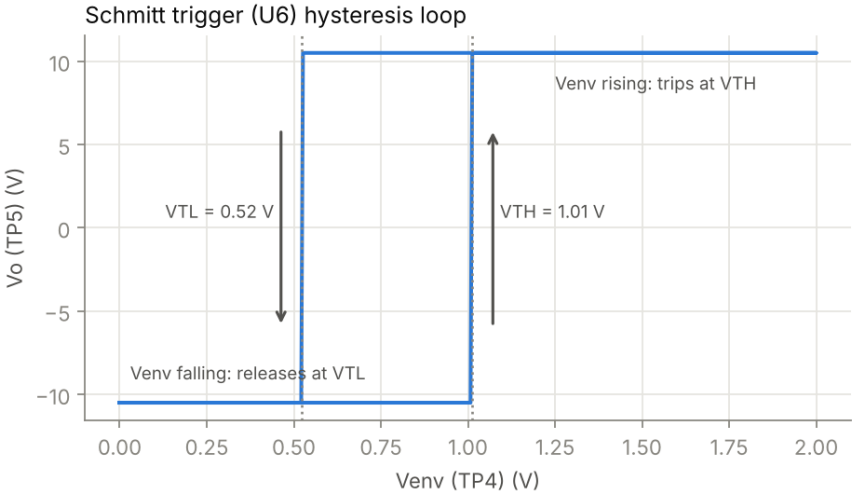


Figure 11: Hysteresis loop of the comparator: V_o versus V_{env}.

Regenerative feedback. When the comparator trips, the term $V_o\,k$ changes from $-0.244\,\mathrm{V}$ to $+0.244\,\mathrm{V}$ and V_+ jumps from $0.75\,\mathrm{V}$ to $1.228\,\mathrm{V}$, far away from V_{ref}. This is why noise cannot switch the output back. The current through R11 and R10 (about $15\,\mu\mathrm{A}$) is supplied by buffer U5, so it does not disturb the discharge of C1.

Sensitivity. A one-volt change in V_{sat} shifts each threshold by $23\,\mathrm{mV}$; a $\pm 5\,\%$ error on the $+12\,\mathrm{V}$ rail shifts them by $\pm 38\,\mathrm{mV}$.

Hand calculations. Figure 12 shows the hand calculation of the thresholds, of R_{11} from the hysteresis, of the R_{12}–R_{13} divider, and the check of V_{TH} and V_{TL}.

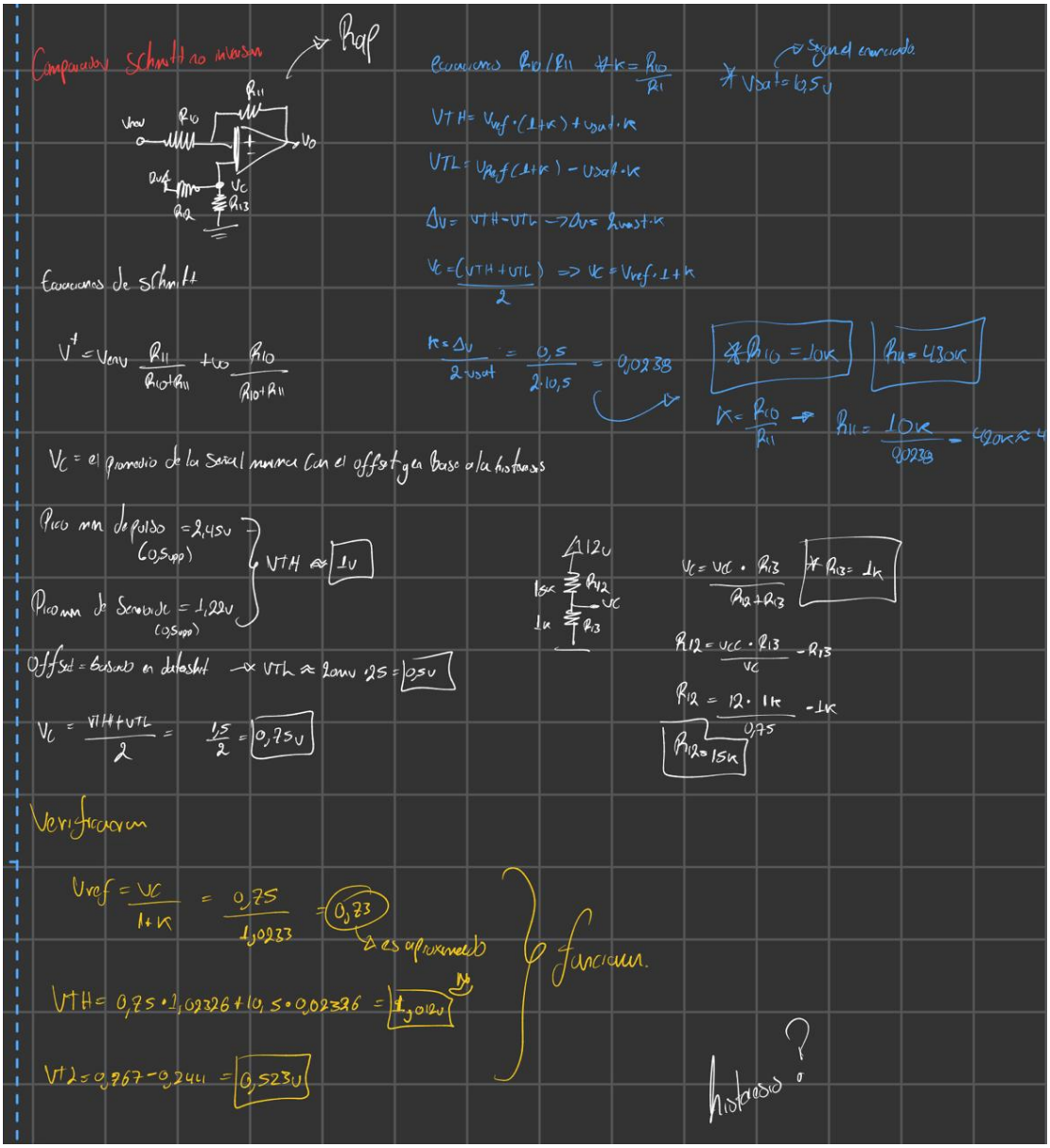


Figure 12: Hand calculations: non-inverting Schmitt comparator (U6) and reference divider (handwritten, in Spanish).

6.8 Stage 5: LED indicator

With $V_o = +10.5\,\text{V}$, D5 and the LED conduct:

$$I_{LED} = \frac{V_{sat} - V_{D5} - V_{LED}}{R_{14}} = \frac{10.5 - 0.7 - 2.0}{1\,\text{k}\Omega} = 7.8\,\text{mA} \tag{27}$$

The D5 drop (0.7 V) is that of a silicon diode at mA-level current, and the drop of the red LED (2 V) is taken from its datasheet. With $V_o = -10.5\,\text{V}$, D5 blocks the reverse voltage (the 1N4148 withstands 100 V) and protects the LED, which only tolerates about 5 V in reverse.

A blue LED was used in the build, with a drop of about 3 V; with the measured high level of 11.1 V the current is $(11.1 - 0.7 - 3.0)/1\,\mathrm{k\Omega} \approx 7.4\,\mathrm{mA}$, practically the same as in the design.

The heart rate is obtained by counting the rising edges at TP5: $HR = 60/T_{RR}$, where T_{RR} is the time between two edges.

7 Simulation

7.1 Behavioral simulation (Python)

Each stage was coded with the equations of the previous section at 20 000 samples per second (code in Appendix A): a gain of 4.9 with clipping at $\pm 10.5\,\mathrm{V}$, ideal rectification, charging with $\tau_c = 0.1\,\mathrm{ms}$ and exponential discharge with $\tau = 100\,\mathrm{ms}$, and a comparator with state. It does not include slew rate or diode drops, which at these frequencies account for less than 1 %.

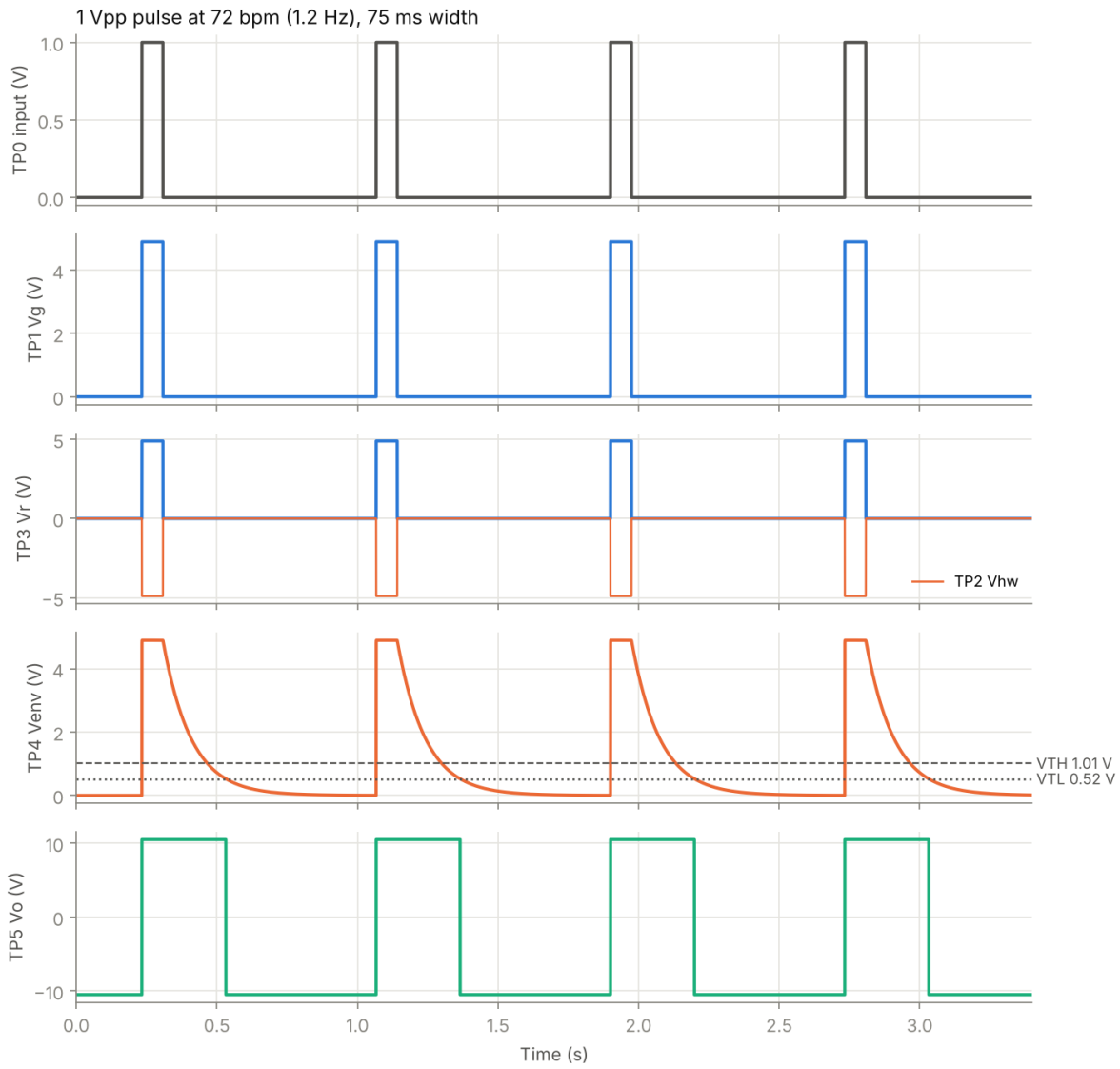


Figure 13: Waveforms at TP0 to TP5 with a $1\,V_{pp}$ pulse at 72 bpm.

Table 10: Rate and amplitude sweep with a pulse (positive and negative polarity gave identical results).

HR (bpm)	Input (V_{pp})	Beats	Detected	Width (ms)	V_{env} peak (V)	V_{env} before (V)
40	0.5	6	6	229	2.45	0.00
40	1.0	6	6	299	4.90	0.00
40	2.0	6	6	368	9.79	0.00
72	0.5	12	12	229	2.45	0.00
72	1.0	12	12	299	4.90	0.00
72	2.0	12	12	368	9.79	0.00
120	0.5	20	20	229	2.45	0.03
120	1.0	20	20	299	4.90	0.07
120	2.0	20	20	368	9.79	0.14

HR (bpm)	Input (V_{pp})	Beats	Detected	Width (ms)	V_{env} peak (V)	V_{env} before (V)
150	0.5	25	25	229	2.45	0.09
150	1.0	25	25	299	4.90	0.19
150	2.0	25	25	368	9.79	0.38
180	0.5	30	30	229	2.45	0.18
180	1.0	30	30	299	4.90	0.37
180	2.0	30	1 (stuck high)	—	9.79	0.74

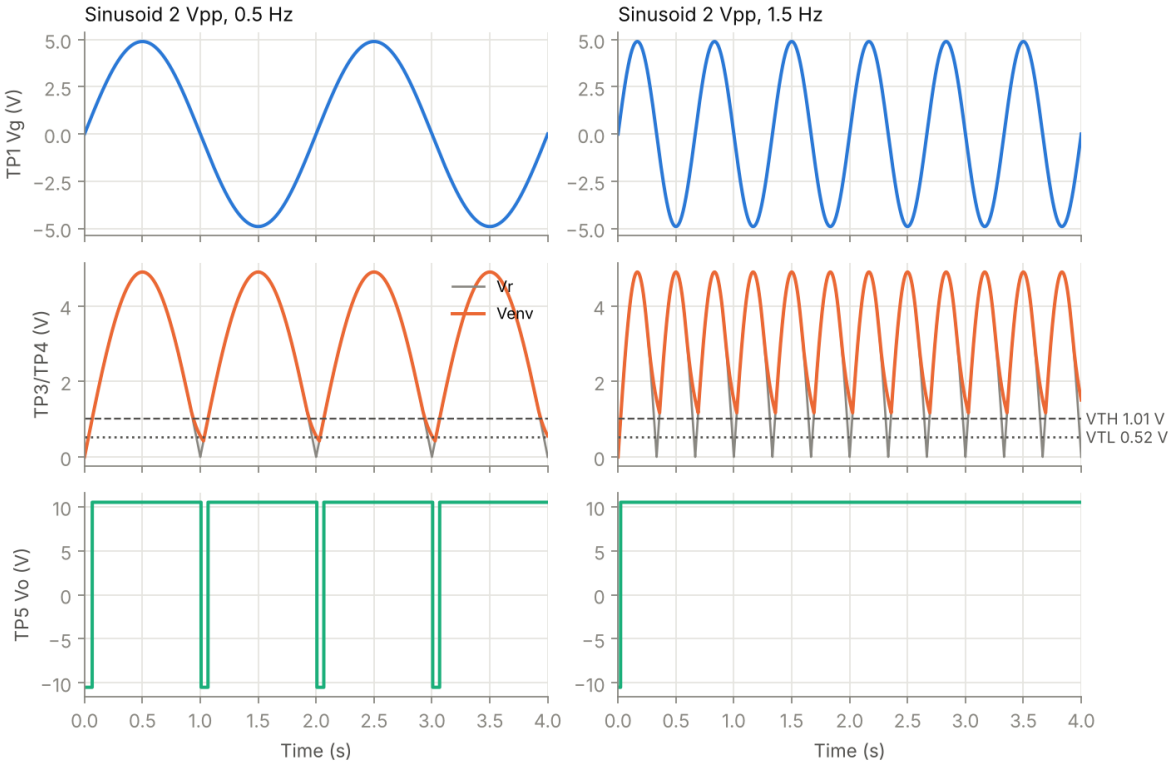


Figure 14: $2\,V_{pp}$ sine wave at 0.5 Hz (two pulses per cycle) and at 1.5 Hz (output stuck high).

Table 11: Results with a sine wave (minimum of V_{env} in parentheses).

Frequency	$0.5\,V_{pp}$	$1\,V_{pp}$	$2\,V_{pp}$
0.5 Hz	2 pulses/cycle (0.11 V)	2 pulses/cycle (0.21 V)	2 pulses/cycle (0.42 V)
1.0 Hz	2 pulses/cycle (0.20 V)	2 pulses/cycle (0.41 V)	Stuck high (0.82 V)
1.5 Hz	2 pulses/cycle (0.29 V)	Stuck high (0.58 V)	Stuck high (1.17 V)

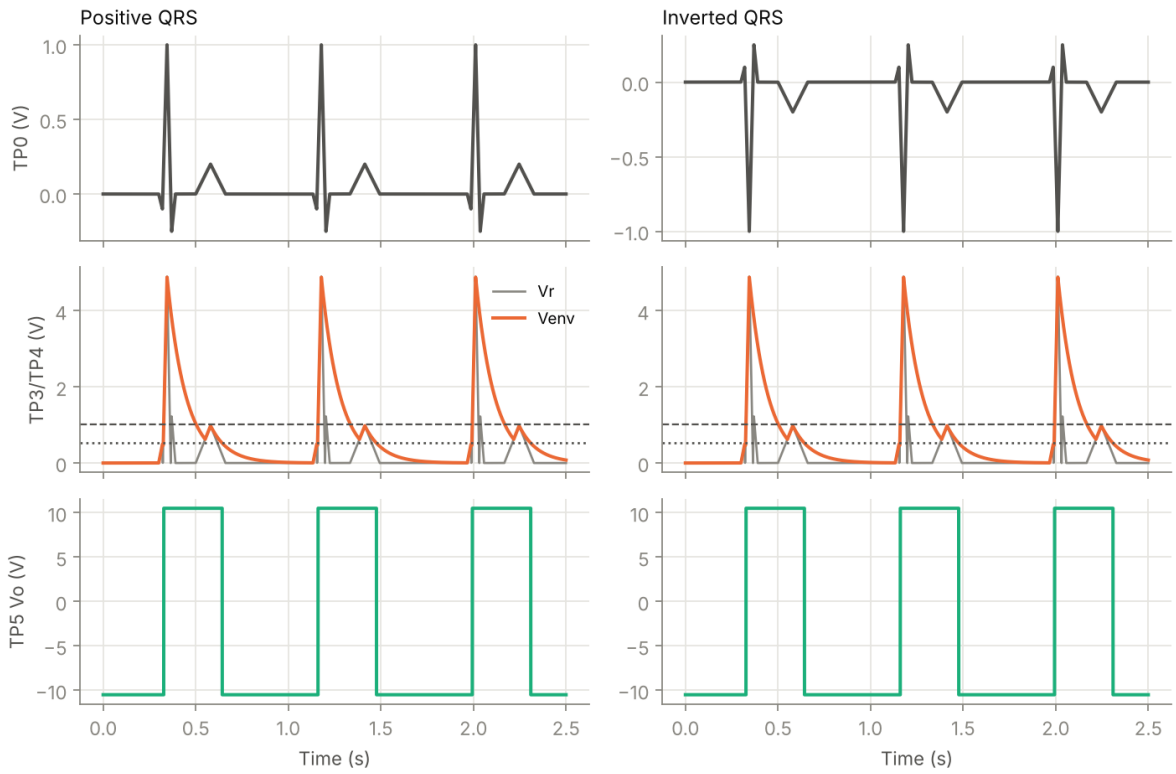


Figure 15: Synthetic heartbeat with positive and inverted QRS: a single pulse per beat in both cases.

With a synthetic heartbeat (a 1 V QRS with negative Q and S waves and a 0.2 V T wave) the circuit gives one pulse per beat for both polarities. After the gain, the T wave reaches 0.98 V, just below V_{TH}, and it arrives while V_{env} is still above V_{TL}, so it only lengthens the pulse.

7.2 Proteus simulation

Op-amp model. The LM741 model was used in Proteus (one amplifier per IC, pin 7 to +12 V and pin 4 to −12 V). At the frequencies of this circuit its relevant parameters (slew rate of 0.5 V/µs, $V_{sat} \approx \pm 10$ V, bias current of about 80 nA) are equivalent to those of the LM324 and TL074 used in the build.

Simulated schematic. Figure 16 shows the circuit built in Proteus. The IC designators were assigned by the program and do not match those in this report; the correspondence is given in Table 12. Channel D of the oscilloscope is connected after D5 (LED node), which is why the low level of the output reads 0 V instead of $-V_{sat}$ in the screenshots.

Table 12: Correspondence between the report numbering and the Proteus numbering.

Function	Report	Proteus
Amplifier	U1	U1
Half-wave	U2	U5
Summer	U3	U4
Detector	U4	U9
Buffer	U5	U2
Schmitt	U6	U7
LED	LED1	D6 (LED-AQUA)

Test sources.

- **PULSE:** low level 0 V; high level 0.5, 1 or 2 V (−1 V for negative polarity); 10 µs edges; width of 9 % of the period or a fixed 75 ms. The result screenshots were taken at 0.5 Hz with 9 % (180 ms).
- **SINE:** offset 0, peak-to-peak amplitude of 0.5 to 2 V, frequency 0.5 Hz (and 1 or 1.5 Hz to show the limit set by τ).
- **FILE:** text file with 12 synthetic heartbeats (time and voltage), with the *Repeat Waveform* option enabled.

Instruments. Virtual oscilloscope with A at the input or TP1, B at TP3, C at TP4 and D at TP5, all DC-coupled; time base of 200 ms/div (500 ms/div for the sine wave) and rising-edge trigger on channel A.

Frequency scaling. If the real-time simulation is slow, all frequencies are multiplied by k and C1 is divided by k (it is the only component that sets timing). With $k = 10$: a 12 Hz pulse, C1 = 100 nF; the measured times are multiplied by 10.

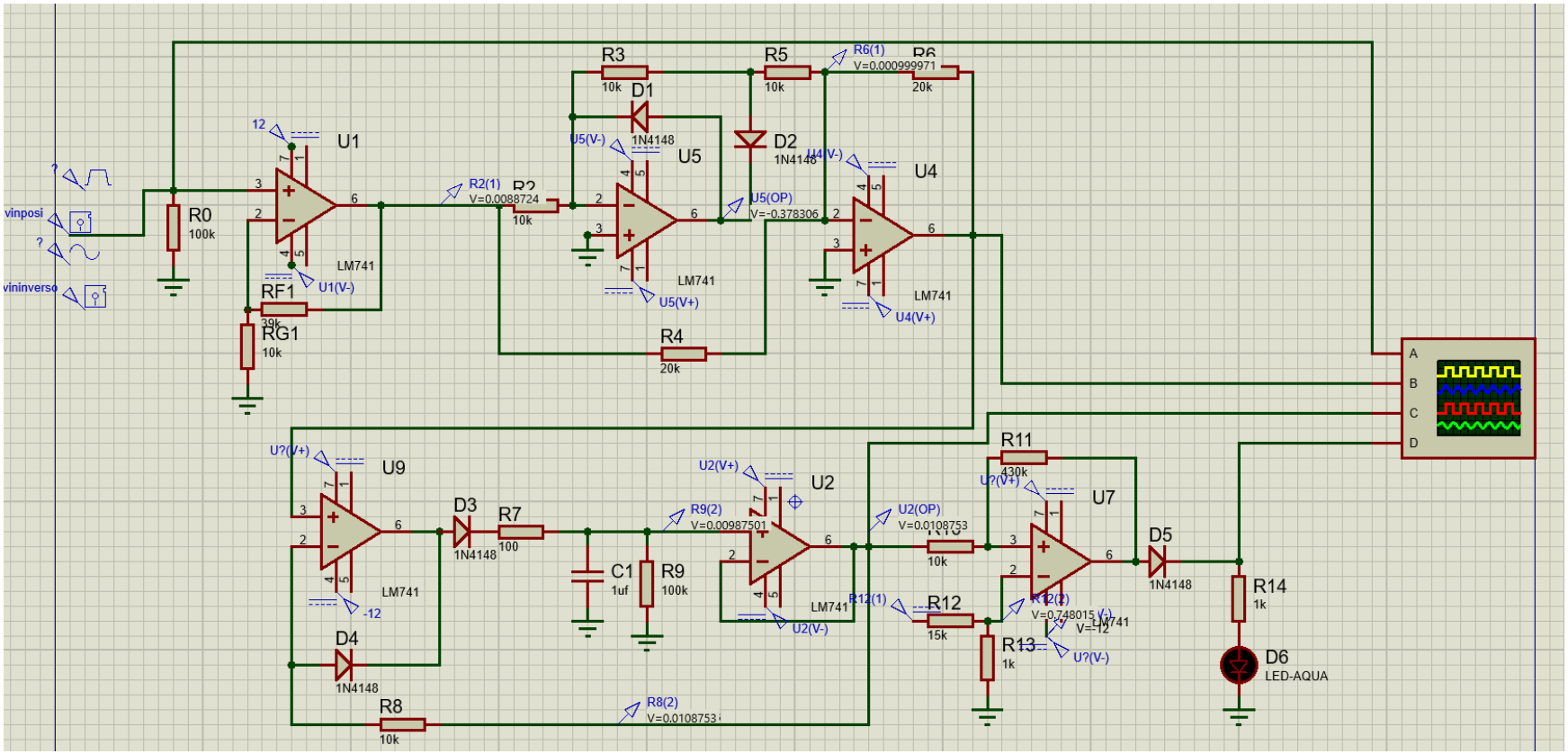


Figure 16: Circuit simulated in Proteus with LM741 op-amps and a virtual oscilloscope (A: input, B: V_r, C: V_{env}, D: output after D5).

8 Experimental Procedure

8.1 Assembly

1. Power the ICs from $\pm 12\,\mathrm{V}$ and place a $100\,\mathrm{nF}$ capacitor between each supply pin and ground.
2. Build and verify one stage at a time: check TP1 before connecting the rectifier, and so on.
3. Use a common ground for the power supply, generator and oscilloscope.
4. Check the orientation of D1 to D4: a reversed diode in the rectifier gives a half-wave output or leaves the output stuck at a rail.
5. The circuit is intended only for tests with a generator; it is not connected to electrodes on a person, since that requires isolation and safety standards (IEC 60601).

8.2 Agilent 33120A generator setup

The initial plan was to use the GW Instek AFG-2025 generator, which only offers pulses and sine waves. In the laboratory it was replaced by the Agilent 33120A because it has a built-in cardiac waveform (CARDIAC) with P, QRS and T waves, much closer to the real signal than the rectangular pulse. The sine wave was generated with the same instrument. The setup was:

1. **Waveform:** `Shift` + `Arb` (*Arb List*), select `CARDIAC` with the arrow keys, `Enter`, and then `Arb` to activate it.
2. **Frequency:** `Freq`, `Enter Number`, `1.2` and the Hz unit arrow key. $1.2\,\mathrm{Hz}$ corresponds to 72 bpm.
3. **Amplitude:** `Ampl`, with 1, 2 and $4\,\mathrm{V_{pp}}$ in the different tests.
4. **High-impedance output termination** (`OUT TERM` = `HIGH Z` in the system menu): by default the instrument assumes a $50\,\Omega$ load and, since the circuit input is $100\,\mathrm{k\Omega}$, it would deliver twice the displayed value.
5. **Offset:** adjusted while watching the input on the oscilloscope with DC coupling, until the baseline between beats sat at $0\,\mathrm{V}$. If the baseline is negative, the rectifier turns it into a constant positive level and the envelope never falls below V_{TL}.

8.3 Measurement points

Table 13: Test points.

TP	Node	What to measure
TP0	Input	Actual V_{pp}, frequency and pulse width
TP1	U1 output (V_g)	Amplitude and actual gain
TP2	Half-wave (V_{hw})	Only the positive half-cycle of V_g, inverted
TP3	U3 output (V_r)	$\lvert V_g \rvert$, with no step at the zero crossing
TP4	U5 output (V_{env})	Peak and τ (time to 37 % of the peak)
TP5	U6 output	$\pm V_{sat}$ levels, pulse width and frequency

With a two-channel oscilloscope, measurements are taken in consecutive pairs (TP0–TP1, TP1–TP3, TP3–TP4, TP4–TP5) triggering on channel 1. The thresholds are measured in XY mode (X = TP4, Y = TP5) with a slow ramp at the Schmitt input. The offset is measured at TP4 with the input at 0 V.

9 Results

9.1 Proteus simulation

1 V pulse at 0.5 Hz. With a 9 % duty cycle the pulse lasts $w = 0.09 \times 2\,\text{s} = 180\,\text{ms}$. The expected output width is

$$w + \tau \ln \frac{V_p}{V_{TL}} = 180\,\text{ms} + 100\,\text{ms} \cdot \ln \frac{4.95}{0.523} = 180\,\text{ms} + 225\,\text{ms} = 405\,\text{ms} \tag{28}$$

In the screenshot (Figure 17) the output stays high for about 400 ms, V_r reaches 4.90 V and V_{env} 4.95 V; the envelope stays flat during the pulse and then decays exponentially.

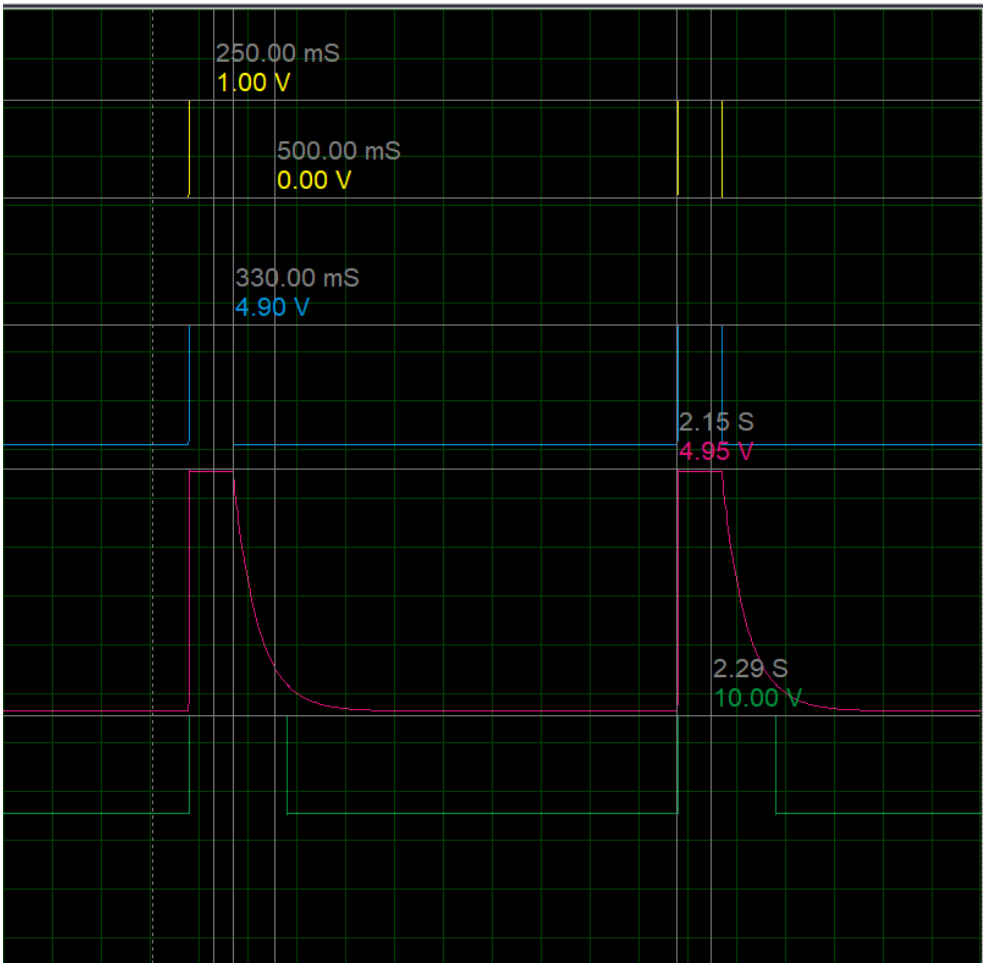


Figure 17: Proteus, 1 V pulse at 0.5 Hz. Yellow: input; blue: V_r (TP3); pink: V_{env} (TP4); green: output after D5.

2 V_{pp} sine wave at 0.5 Hz. V_r shows the two positive humps per cycle with a 4.90 V peak, which confirms full-wave rectification. The output is high almost all the time and drops briefly near each zero crossing, when V_{env} falls below V_{TL}: there are two rising edges per cycle, one per half-cycle (Figure 18).

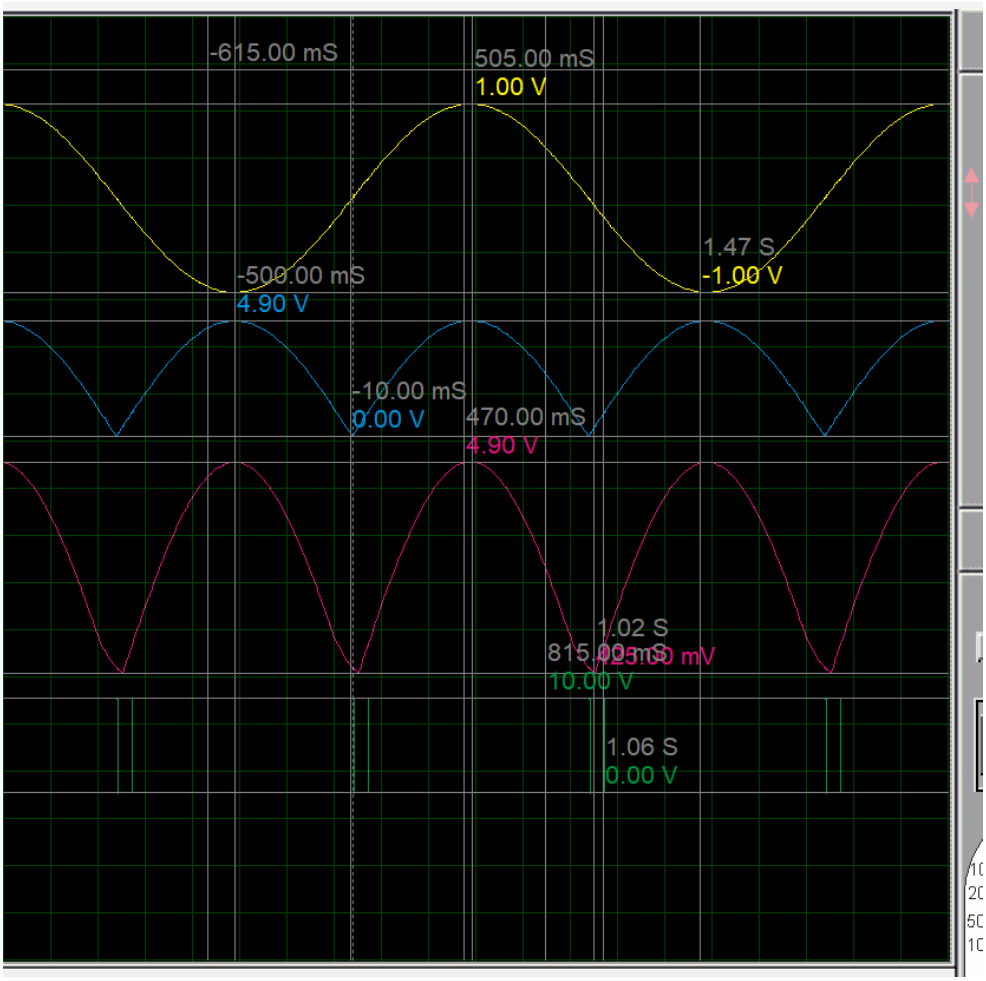


Figure 18: Proteus, $2\,V_{pp}$ sine wave at 0.5 Hz: two output pulses per cycle.

Synthetic heartbeat (FILE generator). With the 12-beat file (1 V QRS, negative Q and S waves and a 0.2 V T wave), V_r shows the rectified lobes and the T wave, but V_{env} merges them into a single hump per beat. The output gives one pulse of about 325 ms per beat, and the T wave causes no false triggers (Figure 19).

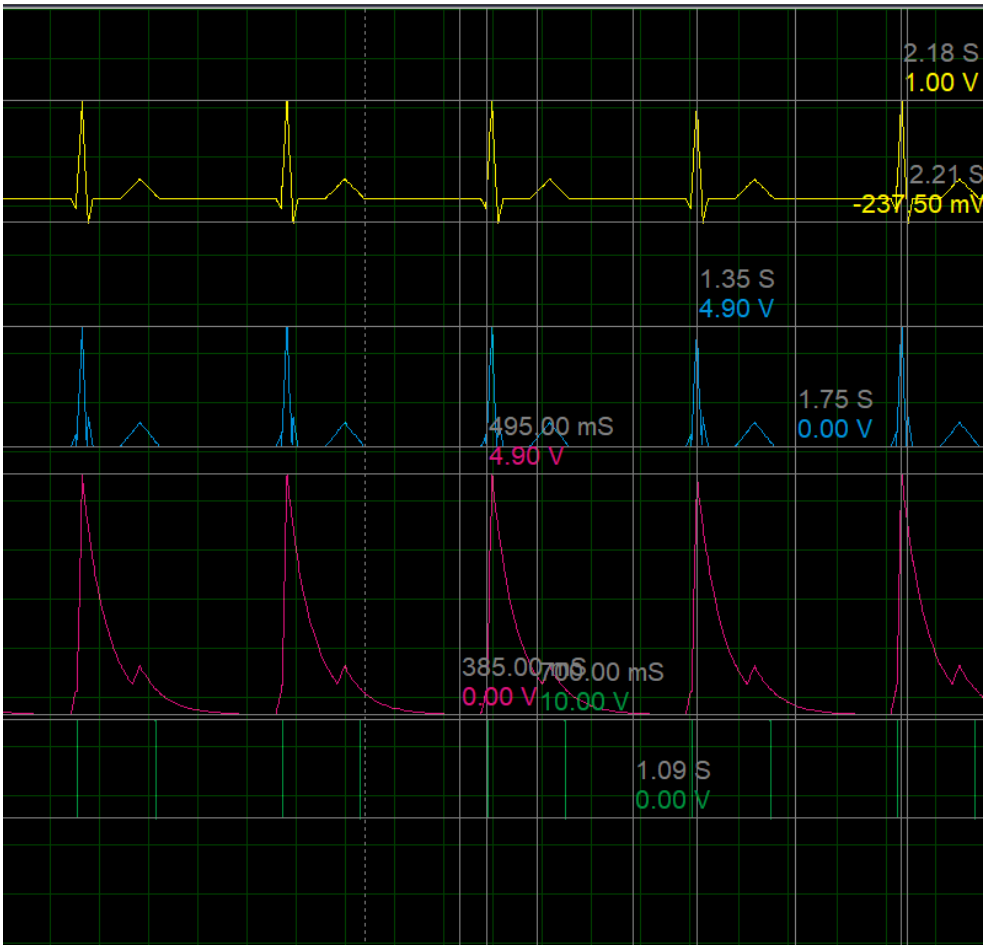


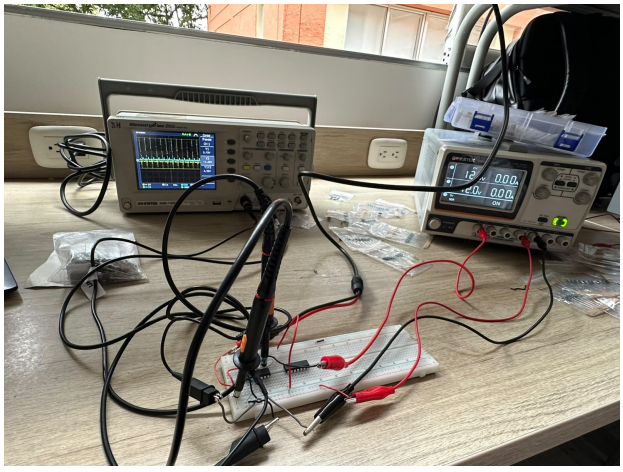
Figure 19: Proteus, synthetic heartbeat at 72 bpm: one output pulse per beat.

Table 14: Comparison of theoretical values, Proteus simulation and laboratory measurements.

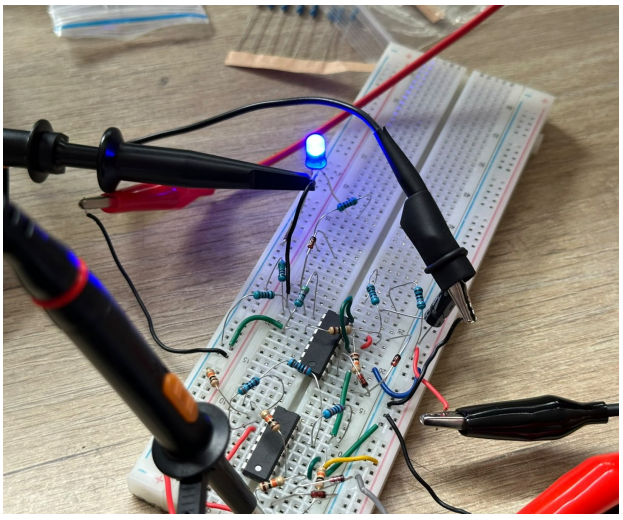
Quantity	Theoretical	Proteus	Error (%)	Measured
1 V *pulse,* 0.5 Hz, 180 ms *wide*				
Gain V_r/V_{in}	4.90	4.90	0.0	
Peak of V_r (TP3)	4.90 V	4.90 V	0.0	
Peak of V_{env} (TP4)	4.90 V	4.95 V	1.0	
Output pulse width	405 ms	$\approx$ 400 ms	1.2	
Output period	2 s (30 bpm)	2 s	0.0	
$2\,V_{pp}$ *sine wave,* 0.5 Hz				
Peak of V_r	4.90 V	4.90 V	0.0	
Pulses per cycle	2	2	—	
Synthetic heartbeat, 72 bpm				
Pulses per beat	1	1	—	
Output pulse width	$\approx$ 320 ms	$\approx$ 325 ms	1.6	
Additional laboratory measurements				
Discharge τ	100 ms			
V_{TH}	1.01 V			
V_{TL}	0.52 V			
Offset at TP4 (input at 0 V)	$<$ 50 mV			

9.2 Laboratory build

The circuit was built on a breadboard with two LM324N chips powered from $\pm$12 V by a dual supply in series mode, with a common ground for the supply, generator and oscilloscope (Figure 20).



(a) Test bench: oscilloscope, dual supply and breadboard.

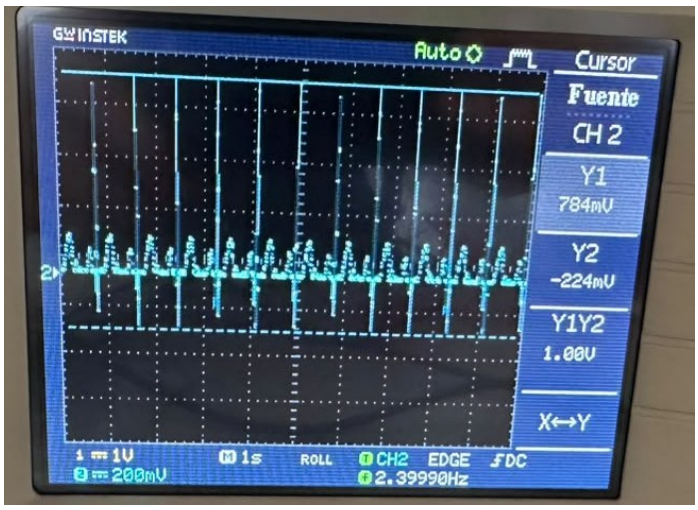


(b) Build with two LM324N chips; LED lit during a beat.

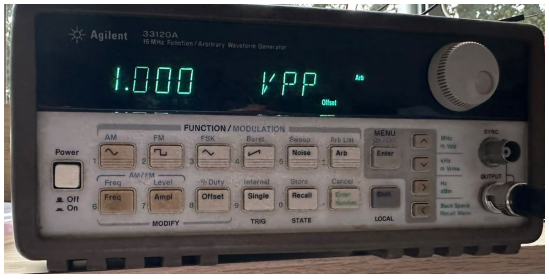
Figure 20: The detector built in the laboratory.

9.2.1 Input signal

Figure 21 shows the $1\,V_{pp}$ CARDIAC waveform measured at the input with DC coupling. The P wave, the QRS and the T wave can be distinguished, with the baseline at 0 V. The R peak reaches 0.78 V and the S wave -0.22 V, with 1.00 V between cursors, equal to the value programmed on the generator.



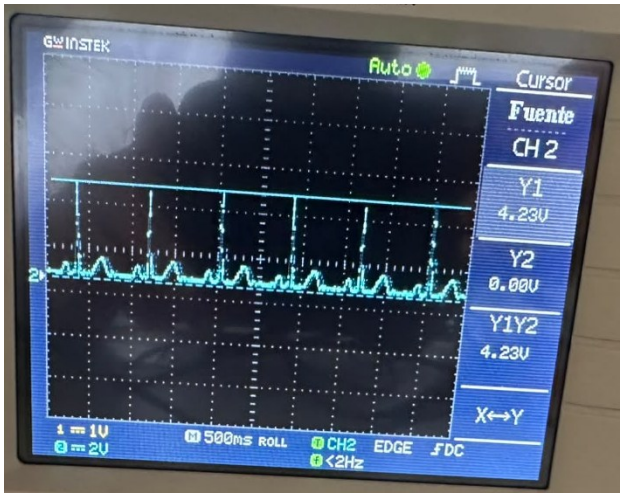
(a) Input (200 mV/div, 1 s/div).



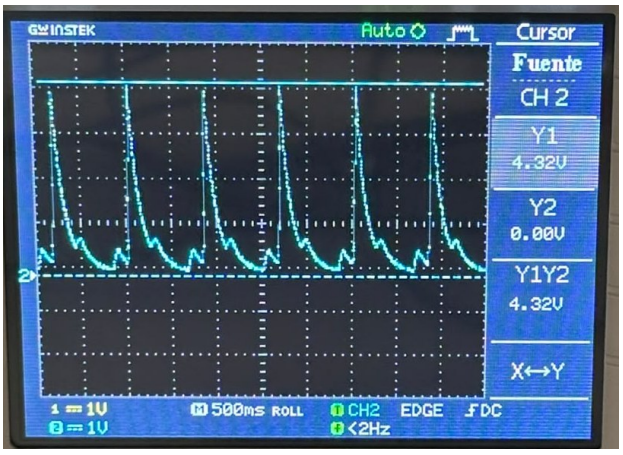
(b) Generator set to $1.000\,V_{pp}$, *Arb* waveform.

Figure 21: Cardiac test signal from the Agilent 33120A.

9.2.2 Stage-by-stage response to the cardiac signal ($1\,V_{pp}$, 1.2 Hz)



(a) TP3, V_r (2 V/div, 500 ms/div).



(b) TP4, V_{env} (1 V/div, 500 ms/div).

Figure 22: Rectifier and peak detector outputs with the cardiac signal.

- **Rectifier (TP3):** the whole signal becomes positive and the baseline stays at 0 V, with no step at the zero crossing. The rectified R peak reaches 4.23 V. With the design gain and the

input peak read in Figure 21, $4.9 \times 0.78 = 3.84\,\text{V}$ was expected. The difference, about 10 %, is attributed mainly to the fact that at 1 s/div in *roll* mode the oscilloscope takes few samples per division and clips the R peak, which lasts a few tens of milliseconds. The tolerance of Rf1 and Rg1 (5 %) also contributes.

- **Peak detector (TP4):** the envelope rises with the QRS up to 4.32 V and discharges exponentially. The 0.09 V difference with respect to V_r is the accumulated offset of U4 and U5, of the same order as observed in Proteus. The rectified T wave appears as a small hump during the discharge.

9.2.3 Comparator output

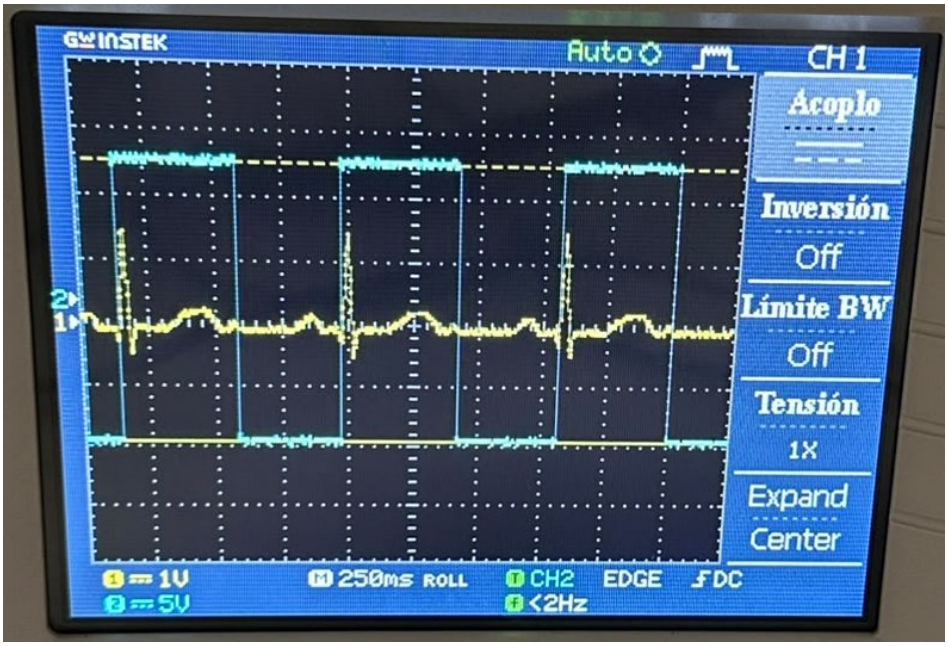


Figure 23: Cardiac input (CH1, yellow, 1 V/div) and Schmitt output at TP5 (CH2, blue, 5 V/div), at 250 ms/div: one rectangular pulse for each QRS.

Figure 23 is the main result of the build: each QRS at the input produces a single rising edge at the output, and the T wave does not generate extra triggers. The output switches between the saturation levels of the LM324N. As measured with the oscilloscope (Figure 25a), the high level is 11.1 V and the output frequency is 1.199 Hz, equal to that of the input. With this high level, the actual thresholds are $V_{TH} = 0.767 + 11.1\,(0.02326) = 1.03\,\text{V}$ and $V_{TL} = 0.51\,\text{V}$, within 20 mV of the design values. Measured after D5, across the LED, the output goes from 0 V to 9.9 V (Figure 24): D5 blocks the negative level, as designed.

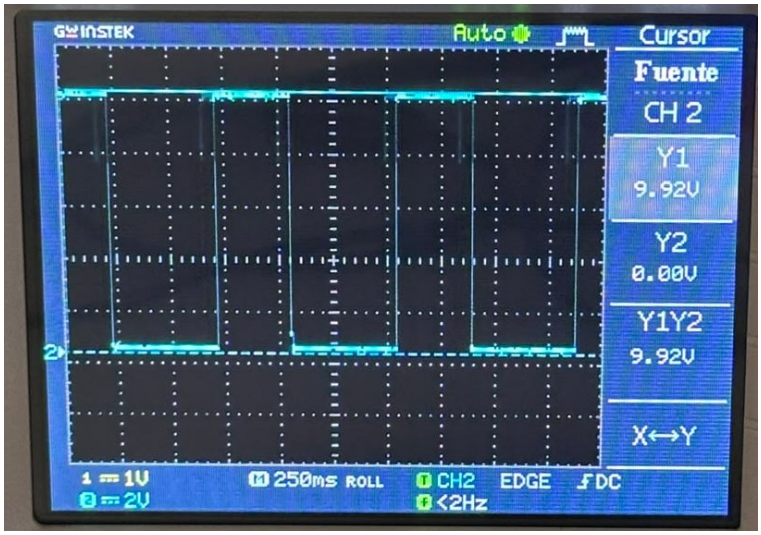
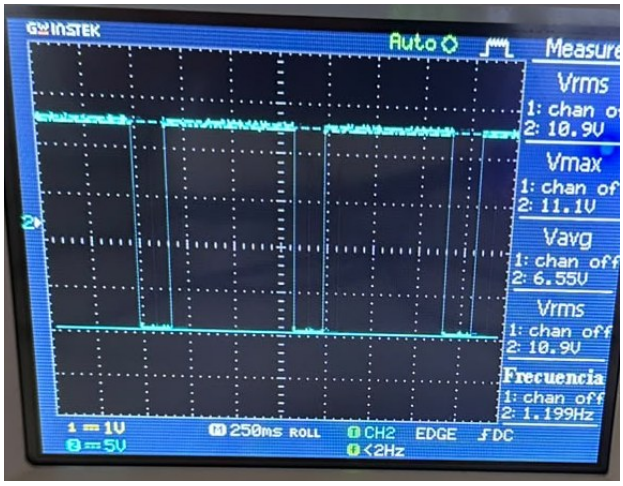


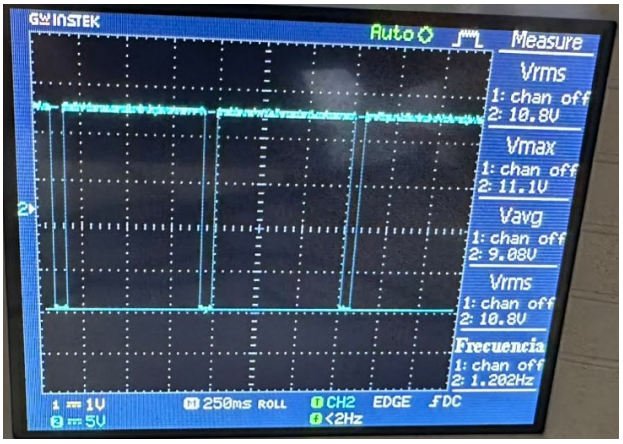
Figure 24: Output measured after D5, across the LED (2 V/div, 250 ms/div): from 0 V to 9.92 V.

9.2.4 Test with $4\,V_{pp}$ at 1.2 Hz

To check the robustness of the design, the amplitude was increased to 2 and then to $4\,V_{pp}$, twice the specified maximum. At 1.2 Hz the circuit kept delivering one pulse per beat, with 1.202 Hz measured at the output (Figure 25b).

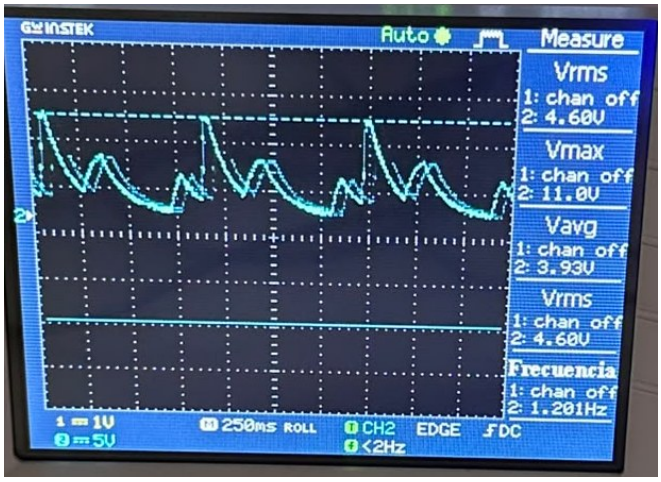


(a) $1\,V_{pp}$: $V_{max} = 11.1\,V$, $V_{avg} = 6.55\,V$, 1.199 Hz.

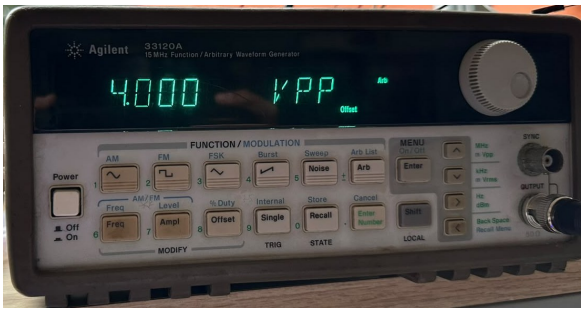


(b) $4\,V_{pp}$: $V_{max} = 11.1\,V$, $V_{avg} = 9.08\,V$, 1.202 Hz.

Figure 25: Automatic measurements of the Schmitt output (TP5, 5 V/div, 250 ms/div).



(a) V_{env} with $4\,\mathrm{V_{pp}}$: saturates at $11.0\,\mathrm{V}$.



(b) Generator set to $4.000\,\mathrm{V_{pp}}$.

Figure 26: Out-of-specification test: $4\,\mathrm{V_{pp}}$ input at $1.2\,\mathrm{Hz}$.

Since the output alternates between $+V_H \approx 11.1\,\mathrm{V}$ and $-V_L \approx 10.5\,\mathrm{V}$, the duty cycle d is obtained from the measured average value, $V_{avg} = d\,V_H - (1-d)\,V_L$:

$$d = \frac{V_{avg} + V_L}{V_H + V_L} \tag{29}$$

Table 15: Comparator output with the cardiac signal at $1.2\,\mathrm{Hz}$ ($T \approx 833\,\mathrm{ms}$).

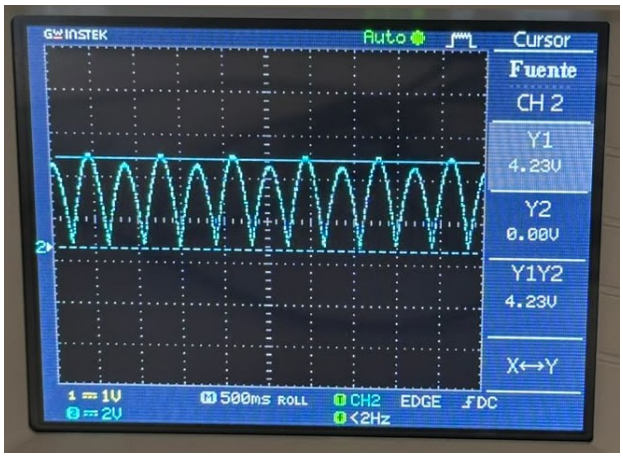
Input	V_{env} peak	f_{out}	V_{avg}	d	Time high	Time low
$1\,\mathrm{V_{pp}}$	$4.32\,\mathrm{V}$	$1.199\,\mathrm{Hz}$	$6.55\,\mathrm{V}$	$79\,\%$	$658\,\mathrm{ms}$	$176\,\mathrm{ms}$
$4\,\mathrm{V_{pp}}$	$11.0\,\mathrm{V}$ (saturated)	$1.202\,\mathrm{Hz}$	$9.08\,\mathrm{V}$	$91\,\%$	$754\,\mathrm{ms}$	$78\,\mathrm{ms}$

Two effects explain why the circuit tolerates $4\,\mathrm{V_{pp}}$ at 72 bpm:

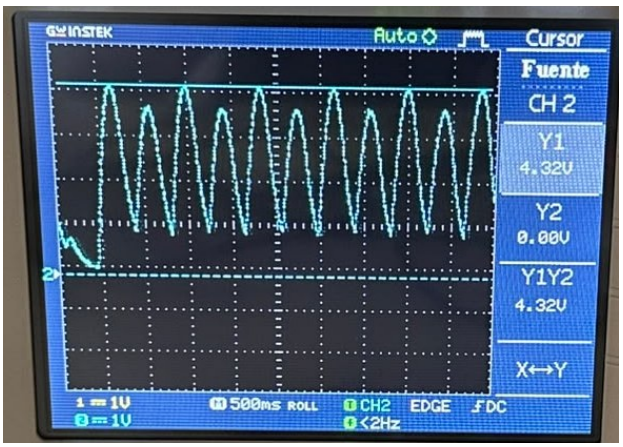
- **Saturation limits the envelope.** With $4\,\mathrm{V_{pp}}$, U1 saturates and V_{env} does not exceed about $11\,\mathrm{V}$. The discharge time is bounded to $\tau \ln(11/0.52) \approx 305\,\mathrm{ms}$ even if the input keeps growing, and the comparator manages to release within the $833\,\mathrm{ms}$ period.
- **The T wave lengthens the pulse.** With the cardiac waveform the output pulse lasts considerably longer than with the $75\,\mathrm{ms}$ rectangular pulse ($658\,\mathrm{ms}$ versus about $300\,\mathrm{ms}$), because the rectified T wave recharges C1 and keeps V_{env} above V_{TL}. With $4\,\mathrm{V_{pp}}$ the amplified T wave is larger and the time low shrinks to $78\,\mathrm{ms}$: the circuit still works, but close to its limit. At higher rates or amplitudes the beats would start to merge.

When U1 saturates, the amplitude information is lost, but detection does not depend on it: it is enough for V_{env} to exceed V_{TH} and fall back below V_{TL} in every period.

9.2.5 Sine-wave test



(a) TP3, V_r: full wave, 4.23 V peak.



(b) TP4, V_{env}: oscillates between ≈ 1.6 V and 4.32 V.

Figure 27: Response to the sine wave (500 ms/div).

The rectifier delivers a clean full-wave signal, with two positive half-cycles per cycle and valleys that reach exactly 0 V. This confirms on the actual build that the precision rectifier has no noticeable dead zone. The envelope, on the other hand, never drops below ≈ 1.6 V: the sine wave rises again before C1 discharges down to V_{TL}, so the comparator stays high and does not deliver separate pulses. This is the behavior anticipated by the sine-wave simulation: this type of signal has a much lower frequency limit than the pulse, because it produces two events per cycle and almost never sits at zero.

9.2.6 Summary: calculation, simulation and build

Table 16: Comparison of results.

Quantity	Calculation / Proteus	Build	Comment
Output high level	10.5 V (assumed)	11.1 V	LM324N at ±12 V with a light load
V_{TH} / V_{TL}	1.01 / 0.52 V	1.03 / 0.51 V	Recalculated with the measured high level
Offset $V_{env} - V_r$	0.05 V (Proteus)	0.09 V	Offset of U4 and U5
V_r valley with sine wave	0 V	0 V	No dead zone
Pulses per beat	1	1	No triggers from the T wave
Output frequency	1.2 Hz	1.199 Hz	Equal to the input
Maximum amplitude tested	2 V_{pp} (design)	4 V_{pp}	Works thanks to U1 saturation

10 Analysis and Discussion

10.1 Frequency limits

The upper limit is set by the condition $w + t_{desc} < T$ (equation 19): with the fixed 75 ms pulse width it is 262, 201 and 163 bpm for 0.5, 1 and 2 V_{pp}. It decreases with amplitude because a larger peak takes longer to discharge: doubling the peak adds $\tau \ln 2 \approx 69$ ms. There is no real lower limit, because the circuit is DC-coupled; the pulse only needs to last at least ≈ 0.5 ms to charge C1. The sine wave has a lower limit than the pulse because it produces two events per cycle and the signal is almost never at zero.

10.2 Op-amp comparison

Table 17: Op-amps considered.

Parameter	TL074	LM324 / LM741	Effect on the circuit
Bias current	≈ 30 pA	45–80 nA	Error on C1 below 0.2 % in both cases
Slew rate	13 V/µs	0.5 V/µs	µs transitions: negligible
Output at ± 12 V	± 10.5 V	$\approx \pm 10$ to 11.5 V	Shifts the thresholds by a few mV
Offset on V_{env}	15–20 mV	20–45 mV typical	Well below V_{TL}

10.3 Limitations

- Fixed thresholds: below ≈ 0.4 V_{pp} there is no detection, and above ≈ 2.1 V_{pp} U1 saturates. A real monitor uses an adaptive threshold (a fraction of the previous peak).
- A fixed τ limits the maximum rate to ≈ 163 bpm at 2 V_{pp} for the rectangular pulse. With a cardiac waveform the limit is lower, because the T wave prolongs the envelope: in the build, with 4 V_{pp} at 72 bpm, the comparator was low for only 78 ms between beats.
- No filtering: with a real signal, a tall T wave or 60 Hz noise could trigger the comparator. A real monitor places an instrumentation amplifier and a 0.5 to 40 Hz band-pass filter in front.
- With the 741, the worst-case bias current through R0 can bring the offset to ≈ 0.27 V; if this is a problem, R0 can be lowered to 10 kΩ without affecting the rest of the circuit.

11 Conclusions

- A heartbeat detector that delivers one pulse for every QRS complex was designed using only nonlinear op-amp blocks. In Proteus the values matched the calculated ones with errors of 1.6% or less: peaks of 4.90 V at V_r and 4.95 V at V_{env}, and an output pulse of about 400 ms versus the predicted 405 ms.

- The build with two LM324N chips met the objective with a realistic cardiac signal (CARDIAC waveform of the Agilent 33120A): one rectangular pulse per beat at 1.199 Hz, with no triggers from the T wave. The actual high level was 11.1 V instead of 10.5 V, which shifts the thresholds by less than 20 mV. With 4 V_{pp} at 1.2 Hz, twice the design maximum, the circuit kept detecting every beat, because op-amp saturation limits the envelope to about 11 V and bounds the discharge time. However, the time low dropped to 78 ms: with large amplitudes and a pronounced T wave the design operates close to its limit.

- The full-wave precision rectifier solved the polarity problem: the sine wave gave two pulses per cycle in Proteus, the inverted synthetic heartbeat was detected just like the positive one and, in the build, the full-wave signal reached exactly 0 V at every valley. Placing the diode inside the feedback loop reduces the dead zone from 0.6 V to a few µV; a passive rectifier would have lost 25 % of the smallest signal.

- The summer shows how a linear operation produces a nonlinear function: combining the full signal (weight 1) with the half-wave signal (weight 2) gives $|V_g|$. The analysis also showed that the positive half-cycle depends on a subtraction $(2 - 1)$, so the resistors must be 1 %.

- The discharge time constant $\tau = R_9 C_1$ is the critical parameter. It was not chosen arbitrarily: the 90 to 110 ms window comes from two opposing conditions, merging the QRS lobes and releasing the comparator before the next beat. The formula $HR_{max} = 60/(w + \tau \ln(V_p/V_{TL}))$ correctly predicted the only failure case in the simulation (180 bpm at 2 V_{pp}).

- The buffer and resistor R8 are essential: the buffer prevents R8 and the comparator from discharging the capacitor (without it τ would drop to about 9 ms), and R8 allows the buffer output and the virtual ground of the detector to be at different voltages during the discharge.

- The positive feedback of the Schmitt comparator gives two thresholds (1.01 V and 0.52 V) and regenerative transitions: when it trips, the non-inverting input jumps far from the reference and noise cannot switch the output back. The T wave of the synthetic heartbeat caused no false triggers.

- The design has clear limits: fixed thresholds (inputs of 0.4 to 2.1 V_{pp}), a maximum rate of about 160 bpm at the largest amplitude, and no filtering. A next step would be to add an instrumentation amplifier with a 0.5 to 40 Hz band-pass filter in front, use an adaptive threshold, and count the pulses with a microcontroller to display the heart rate.

References

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[5] J. Pan and W. J. Tompkins, "A Real-Time QRS Detection Algorithm", *IEEE Transactions on Biomedical Engineering*, vol. BME-32, no. 3, pp. 230–236, 1985.

[6] J. G. Webster (ed.), *Medical Instrumentation: Application and Design*, Wiley.

[7] Texas Instruments, *TL07xx Low-Noise FET-Input Operational Amplifiers*, datasheet.

[8] Texas Instruments, *LMx24 Quadruple Operational Amplifiers*, datasheet.

[9] Texas Instruments, *LM741 Operational Amplifier*, datasheet.

[10] Vishay / onsemi, *1N4148 Small Signal Fast Switching Diode*, datasheet.

A Behavioral simulation code

The following script generates the figures and tables of the simulation section.

```
"""Behavioral simulation of the analog ECG heartbeat detector.

Models the signal chain stage by stage -- gain stage (TP1), precision
full-wave rectifier (TP2/TP3), peak-envelope detector with R9*C1 discharge
(TP4) and Schmitt-trigger comparator (TP5) -- and evaluates it with
rectangular pulses, sinusoids and a synthetic PWL QRS complex. It also
computes theoretical envelope discharge times / maximum heart rate and a
rectifier resistor-tolerance analysis (worst case + Monte Carlo).

Usage:
    pip install numpy matplotlib
    python sim.py

Figures (f1_rectificador.png ... f7_qrs_pwlin.png) and results.json are
written to the fig/ folder next to this script; a summary is printed.
"""
import numpy as np, matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt, json, os

OUT = os.path.dirname(os.path.abspath(__file__)) + "/fig"
os.makedirs(OUT, exist_ok=True)

# ---------- design parameters ----------
G = 1 + 39e3/10e3            # 4.9
VSAT = 10.5
R7, R9, C1 = 100.0, 100e3, 1e-6
TAU = R9*C1                  # 0.1 s
TAUC = R7*C1                 # 0.1 ms
R10, R11 = 10e3, 430e3
VREF = 12*1e3/(15e3+1e3)     # 0.75
k = R10/R11
VTH = VREF*(1+k) + VSAT*k
VTL = VREF*(1+k) - VSAT*k
FS = 20000.0
DT = 1/FS

BLUE, ORANGE, AQUA, YEL, MAG, VIO = "#2a78d6", "#eb6834", "#1baf7a", "#eda100", "#e87ba4", "#4a3aa7
    "
INK, INK2, MUTED, GRID = "#0b0b0b", "#52514e", "#898781", "#e6e5e0"
plt.rcParams.update({"font.size": 10, "axes.edgecolor": MUTED, "axes.labelcolor": INK2,
                     "xtick.color": MUTED, "ytick.color": MUTED, "axes.spines.top": False,
                     "axes.spines.right": False, "axes.grid": True, "grid.color": GRID,
                     "grid.linewidth": 0.8, "figure.facecolor": "white", "axes.facecolor": "white",
                     "legend.frameon": False, "lines.linewidth": 1.8})
```

```
45
46  def chain(vin):
47      vg = np.clip(G*vin, -VSAT, VSAT)
48      vhw = np.where(vg > 0, -vg, 0.0)
49      vr = np.clip(-(vg + 2*vhw), -VSAT, VSAT)
50      env = np.zeros_like(vr); vo = np.zeros_like(vr)
51      e = 0.0; s = -VSAT
52      dec = np.exp(-DT/TAU); chg = 1-np.exp(-DT/TAUC)
53      for i, x in enumerate(vr):
54          e *= dec
55          if x > e: e += (x-e)*chg
56          env[i] = e
57          if s < 0 and e > VTH: s = VSAT
58          elif s > 0 and e < VTL: s = -VSAT
59          vo[i] = s
60      return vg, vhw, vr, env, vo
61
62  def pulse(t, f, A, w=0.075, pol=1):
63      return pol*A*(((t % (1/f)) < w).astype(float))
64
65  def stats(t, vo, nbeats_expected, f):
66      hi = vo > 0
67      rises = np.where(np.diff(hi.astype(int)) == 1)[0]
68      falls = np.where(np.diff(hi.astype(int)) == -1)[0]
69      widths = []
70      for r in rises:
71          fl = falls[falls > r]
72          if len(fl): widths.append((fl[0]-r)*DT)
73      stuck = hi[-1] and len(falls) < len(rises)-1
74      return len(rises), (np.median(widths)*1e3 if widths else float("nan")), stuck
75
76  # ---------- pulse test-case table ----------
77  rows = []
78  T_SIM = 10.0
79  t = np.arange(0, T_SIM, DT)
80  for bpm in [40, 72, 120, 150, 180]:
81      f = bpm/60
82      for A in [0.5, 1.0, 2.0]:
83          for pol in [1, -1]:
84              vin = pulse(t+0.2, f, A, pol=pol)   # +0.2 s: first pulse not at t=0
85              vg, vhw, vr, env, vo = chain(vin)
86              n_exp = int(np.sum(np.diff((((t+0.2) % (1/f)) < 0.075).astype(int)) == 1))
87              n, wms, stuck = stats(t, vo, n_exp, f)
88              # envelope minimum just before each beat (last complete beat)
89              starts = np.where(np.diff((((t+0.2) % (1/f)) < 0.075).astype(int)) == 1)[0]
90              pre = [env[s0-1] for s0 in starts[1:]]
91              rows.append(dict(bpm=bpm, Vpp=A, pol=pol, expected=n_exp, detected=n,
92                               width_ms=round(wms, 0), peak=round(env.max(), 2),
93                               env_min_before=round(min(pre), 2) if pre else None))
```

```
94
95  # ---------- theoretical vs simulated discharge time ----------
96  theo = []
97  for A in [0.5, 1.0, 2.0]:
98      Vp = min(G*A, VSAT)
99      td = TAU*np.log(Vp/VTL)
100     theo.append(dict(Vpp=A, Vp=round(Vp, 2), t_desc_ms=round(td*1e3, 0),
101                      width_theo_ms=round((0.075+td)*1e3, 0),
102                      fmax_bpm=round(60/(0.075+td), 0)))
103
104 # ---------- sinusoid ----------
105 sine_rows = []
106 for fs_ in [0.5, 1.0, 1.5]:
107     for A in [0.5, 1.0, 2.0]:
108         ts = np.arange(0, 8, DT)
109         vin = (A/2)*np.sin(2*np.pi*fs_*ts)
110         *_, env, vo = chain(vin)
111         n, wms, stuck = stats(ts, vo, 0, fs_)
112         sine_rows.append(dict(f=fs_, Vpp=A, cycles=int(8*fs_), pulses=n,
113                               pulses_per_cycle=round(n/(8*fs_), 2),
114                               env_min=round(env[int(2/DT):].min(), 2)))
115
116 # ---------- rectifier tolerances (Monte Carlo + worst case) ----------
117 def rect_gains(R2, R3, R4, R5, R6):
118     gpos = (R6/R5)*(R3/R2) - R6/R4
119     gneg = R6/R4
120     return gpos, gneg
121 worst = {}
122 for tol in [0.01, 0.05]:
123     best = 0
124     for s2 in [-1, 1]:
125         for s3 in [-1, 1]:
126             for s4 in [-1, 1]:
127                 for s5 in [-1, 1]:
128                     for s6 in [-1, 1]:
129                         gp, gn = rect_gains(10e3*(1+s2*tol), 10e3*(1+s3*tol), 20e3*(1+s4*tol),
130                                             10e3*(1+s5*tol), 20e3*(1+s6*tol))
131                         best = max(best, abs(gp-gn)/gn)
132     rng = np.random.default_rng(1)
133     sd = tol/3
134     R = [10e3, 10e3, 20e3, 10e3, 20e3]
135     m = np.array([R]*20000)*(1+rng.normal(0, sd, (20000, 5)))
136     gp, gn = rect_gains(*m.T)
137     worst[tol] = dict(worst=round(best*100, 1), p95=round(np.percentile(np.abs(gp-gn)/gn, 95)*100,
        2))
138
139 res = dict(G=G, TAU=TAU, VTH=VTH, VTL=VTL, VREF=VREF, k=k, theo=theo, rows=rows,
140            sine=sine_rows, tol=worst)
141 json.dump(res, open(OUT + "/results.json", "w"), indent=1)
```

```
142
143 def thr(ax):
144     ax.axhline(VTH, color=INK2, ls="--", lw=1)
145     ax.axhline(VTL, color=INK2, ls=":", lw=1.2)
146     ax.text(ax.get_xlim()[1], VTH, f" VTH {VTH:.2f} V", va="center", fontsize=8.5, color=INK2)
147     ax.text(ax.get_xlim()[1], VTL, f" VTL {VTL:.2f} V", va="center", fontsize=8.5, color=INK2)
148
149 def save(fig, name):
150     fig.savefig(f"{OUT}/{name}.png", dpi=160, bbox_inches="tight")
151     plt.close(fig)
152
153 # F1 rectifier transfer characteristic
154 vgx = np.linspace(-10, 10, 401)
155 _, vhw, vr, *_ = chain(vgx/G) if False else (None, np.where(vgx > 0, -vgx, 0), np.abs(vgx))
156 fig, ax = plt.subplots(1, 2, figsize=(9, 3.4))
157 ax[0].plot(vgx, np.where(vgx > 0, -vgx, 0), color=ORANGE, label="Vhw (TP2)")
158 ax[0].plot(vgx, np.abs(vgx), color=BLUE, label="Vr (TP3)")
159 ax[0].set_xlabel("Vg (V)"); ax[0].set_ylabel("Output (V)")
160 ax[0].set_title("Transfer curve", fontsize=10.5, color=INK, loc="left")
161 ax[0].legend(loc="lower right")
162 ts = np.arange(0, 2, DT); vin = 0.5*np.sin(2*np.pi*1*ts)
163 vg, vhw2, vr2, *_ = chain(vin)
164 ax[1].plot(ts, vg, color=MUTED, lw=1.2, label="Vg (TP1)")
165 ax[1].plot(ts, vhw2, color=ORANGE, label="Vhw (TP2)")
166 ax[1].plot(ts, vr2, color=BLUE, label="Vr (TP3)")
167 ax[1].set_xlabel("Time (s)"); ax[1].set_ylabel("Voltage (V)")
168 ax[1].set_title("Sinusoid 1 Vpp, 1 Hz", fontsize=10.5, color=INK, loc="left")
169 ax[1].set_ylim(-3.7, 2.9); ax[1].legend(loc="lower center", ncol=3, fontsize=8.5)
170 fig.tight_layout(); save(fig, "f1_rectificador")
171
172 # F2 waveforms, 1 Vpp pulse at 72 bpm
173 ts = np.arange(0, 3.4, DT); f = 1.2
174 vin = pulse(ts+0.6, f, 1.0)
175 vg, vhw, vr, env, vo = chain(vin)
176 fig, ax = plt.subplots(5, 1, figsize=(9, 8.6), sharex=True)
177 for a, y, c, lab in zip(ax, [vin, vg, vr, env, vo], [INK2, BLUE, BLUE, ORANGE, AQUA],
178                         ["TP0 input (V)", "TP1 Vg (V)", "TP3 Vr (V)", "TP4 Venv (V)", "TP5 Vo (V)"
    ]):
179     a.plot(ts, y, color=c); a.set_ylabel(lab)
180 ax[2].plot(ts, vhw, color=ORANGE, lw=1.2, label="TP2 Vhw"); ax[2].legend(loc="lower right",
    fontsize=8.5)
181 ax[3].set_xlim(0, 3.4); thr(ax[3])
182 ax[-1].set_xlabel("Time (s)")
183 ax[0].set_title("1 Vpp pulse at 72 bpm (1.2 Hz), 75 ms width", fontsize=11, color=INK, loc="left")
184 fig.tight_layout(); save(fig, "f2_pulso_72lpm")
185
186 # F3 envelope discharge
187 fig, ax = plt.subplots(figsize=(9, 3.6))
188 tt = np.linspace(0, 0.6, 600)
```

```
for A, c in zip([0.5, 1.0, 2.0], [AQUA, BLUE, VIO]):
    Vp = G*A; v = Vp*np.exp(-tt/TAU); td = TAU*np.log(Vp/VTL)
    ax.plot(tt*1e3, v, color=c, label=f"{A} Vpp (peak {Vp:.2f} V)")
    ax.plot([td*1e3], [VTL], "o", color=c, ms=6, mec="white", mew=1.5)
    ax.annotate(f"{td*1e3:.0f} ms", (td*1e3, VTL), textcoords="offset points", xytext=(4, 8),
                fontsize=8.5, color=INK2)
ax.set_xlim(0, 600); ax.set_ylim(0, 10.5); thr(ax)
ax.set_xlabel("Time since end of pulse (ms)"); ax.set_ylabel("Venv (V)")
ax.set_title("C1 discharge with τ = 100 ms down to VTL", fontsize=11, color=INK, loc="left")
ax.legend(loc="upper right"); fig.tight_layout(); save(fig, "f3_descarga")

# F4 maximum rate vs amplitude and vs tau
fig, ax = plt.subplots(figsize=(9, 3.6))
A = np.linspace(0.25, 2.1, 200)
for tau_, c, lab in [(0.068, AQUA, "τ = 68 ms"), (0.1, BLUE, "τ = 100 ms (design)"), (0.15, ORANGE,
     "τ = 150 ms")]:
    Vp = np.minimum(G*A, VSAT)
    fmax = 60/(0.075 + tau_*np.log(np.maximum(Vp/VTL, 1.0001)))
    ax.plot(A, fmax, color=c, label=lab)
ax.axhline(150, color=INK2, ls="--", lw=1); ax.text(2.1, 150, " 150 bpm", va="center", fontsize
    =8.5, color=INK2)
ax.axvline(VTH/G, color=MUTED, ls=":", lw=1.2)
ax.text(VTH/G, 30, " detection\n threshold", fontsize=8.5, color=INK2)
ax.set_ylim(0, 420); ax.set_xlim(0.2, 2.1)
ax.set_xlabel("Input amplitude (Vpp)"); ax.set_ylabel("Maximum rate (bpm)")
ax.set_title("Maximum heart rate with timely release: 60 / (w + τ·ln(Vp/VTL))", fontsize=10.5, color
    =INK, loc="left")
ax.legend(loc="upper right"); fig.tight_layout(); save(fig, "f4_fmax")

# F5 hysteresis loop
fig, ax = plt.subplots(figsize=(6, 3.6))
x = np.concatenate([np.linspace(0, 2, 400), np.linspace(2, 0, 400)])
s = -VSAT; y = []
for xi in x:
    if s < 0 and xi > VTH: s = VSAT
    elif s > 0 and xi < VTL: s = -VSAT
    y.append(s)
ax.plot(x, y, color=BLUE)
for xv, lab, ha, dx in [(VTL, f"VTL = {VTL:.2f} V", "right", -0.08), (VTH, f"VTH = {VTH:.2f} V", "
    left", 0.08)]:
    ax.axvline(xv, color=MUTED, ls=":", lw=1.2); ax.text(xv+dx, 0.6, lab, fontsize=8.5, color=INK2,
     ha=ha)
ax.annotate("", xy=(VTH+0.06, 6), xytext=(VTH+0.06, -6), arrowprops=dict(arrowstyle="->", color=
    INK2, lw=1.4))
ax.annotate("", xy=(VTL-0.06, -6), xytext=(VTL-0.06, 6), arrowprops=dict(arrowstyle="->", color=
    INK2, lw=1.4)); ax.text(1.25, VSAT-2.2, "Venv rising: trips at VTH", fontsize=8.5, color=INK2);
     ax.text(0.03, -VSAT+1.4, "Venv falling: releases at VTL", fontsize=8.5, color=INK2)
ax.set_xlabel("Venv (TP4) (V)"); ax.set_ylabel("Vo (TP5) (V)")
ax.set_title("Schmitt trigger (U6) hysteresis loop", fontsize=11, color=INK, loc="left")
```

```
fig.tight_layout(); save(fig, "f5_histeresis")

# F6 sinusoid 0.5 Hz and 1.5 Hz, 2 Vpp
fig, ax = plt.subplots(3, 2, figsize=(9, 6), sharex="col")
for j, fs_ in enumerate([0.5, 1.5]):
    ts = np.arange(0, 4, DT); vin = 1.0*np.sin(2*np.pi*fs_*ts)
    vg, vhw, vr, env, vo = chain(vin)
    ax[0, j].plot(ts, vg, color=BLUE); ax[0, j].set_title(f"Sinusoid 2 Vpp, {fs_} Hz", fontsize
    =10.5, color=INK, loc="left")
    ax[1, j].plot(ts, vr, color=MUTED, lw=1.2, label="Vr"); ax[1, j].plot(ts, env, color=ORANGE,
    label="Venv")
    ax[1, j].set_xlim(0, 4); thr(ax[1, j]) if j == 1 else (ax[1, j].axhline(VTH, color=INK2, ls="--
    ", lw=1), ax[1, j].axhline(VTL, color=INK2, ls=":", lw=1.2))
    ax[2, j].plot(ts, vo, color=AQUA); ax[2, j].set_xlabel("Time (s)")
ax[0, 0].set_ylabel("TP1 Vg (V)"); ax[1, 0].set_ylabel("TP3/TP4 (V)"); ax[2, 0].set_ylabel("TP5 Vo
    (V)")
ax[1, 0].legend(loc="upper right", fontsize=8.5)
fig.tight_layout(); save(fig, "f6_sinusoide")

# F7 synthetic PWLIN heartbeat, both polarities
pts = np.array([[0, 0], [0.300, 0], [0.320, -0.10], [0.345, 1.00], [0.370, -0.25], [0.390, 0],
                [0.500, 0], [0.580, 0.20], [0.660, 0], [0.833, 0]])
def pwl(t):
    return np.interp(t % 0.8333, pts[:, 0], pts[:, 1])
fig, ax = plt.subplots(3, 2, figsize=(9, 6), sharex=True)
for j, pol in enumerate([1, -1]):
    ts = np.arange(0, 2.5, DT); vin = pol*pwl(ts)
    vg, vhw, vr, env, vo = chain(vin)
    ax[0, j].plot(ts, vin, color=INK2)
    ax[0, j].set_title("Positive QRS" if pol > 0 else "Inverted QRS", fontsize=10.5, color=INK, loc
    ="left")
    ax[1, j].plot(ts, vr, color=MUTED, lw=1.2, label="Vr"); ax[1, j].plot(ts, env, color=ORANGE,
    label="Venv")
    ax[1, j].axhline(VTH, color=INK2, ls="--", lw=1); ax[1, j].axhline(VTL, color=INK2, ls=":", lw
    =1.2)
    ax[2, j].plot(ts, vo, color=AQUA); ax[2, j].set_xlabel("Time (s)")
ax[0, 0].set_ylabel("TP0 (V)"); ax[1, 0].set_ylabel("TP3/TP4 (V)"); ax[2, 0].set_ylabel("TP5 Vo (V)
    ")
ax[1, 0].legend(loc="upper right", fontsize=8.5)
fig.tight_layout(); save(fig, "f7_qrs_pwlin")

print(json.dumps(dict(G=G, VTH=VTH, VTL=VTL, theo=theo, tol=worst), indent=1))
for r in rows: print(r)
for r in sine_rows: print(r)
```

B Synthetic heartbeat files for Proteus

The files `ecg_beat_positive.txt` and `ecg_beat_inverted.txt` contain 12 heartbeats at 72 bpm (10 s). Each line holds the time in seconds and the voltage in volts. Points of one heartbeat:

Table 18: Points of one synthetic heartbeat.

t (s)	0	0.300	0.320	0.345	0.370	0.390	0.500	0.580	0.660
V (V)	0	0	−0.10	1.00	−0.25	0	0	0.20	0